Diffusion in Silicon

References

- S. Wolf, “Silicon processing for the VLSI era”, Lattice Press, Vol. 1, 2nd Edition.
- S. M. Sze, “VLSI Technology”, 2nd Ed. McGraw Hill, 1988

Outline

- Introduction
- Mathematic Models for Diffusion
- Atomistic Models of Diffusion
- Diffusion of Major Dopants
- Secondary Effects
- Diffusion System

Introduction

- Diffusion in Si: to introduce dopants into Si

- Diffusion process is widely used in semiconductor industry
  Examples:
  - BJT: bases, emitters, and resistors in bipolar technology
  - CMOS: source/drain regions, wells, and polysilicon gate in MOS device technology

- Methods to introduce dopants into silicon
  - Diffusion at high temperature
  - Ion implantation

- Key process parameters:
  - Dopant profile and depth, uniformity, and reproducibility

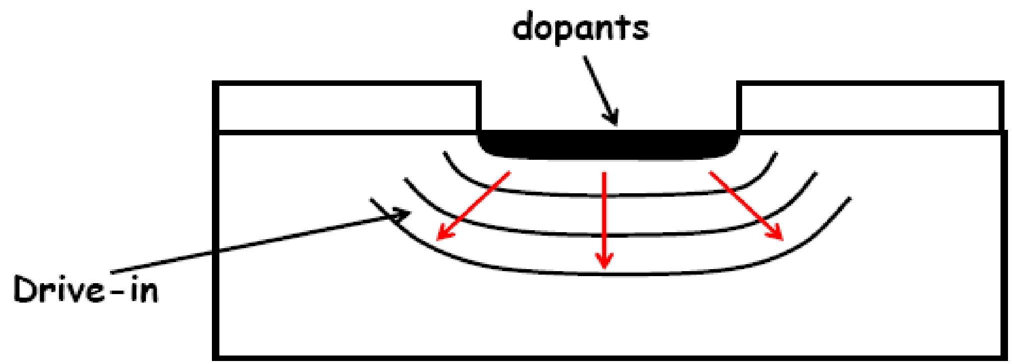
Typical process for dopant introduction

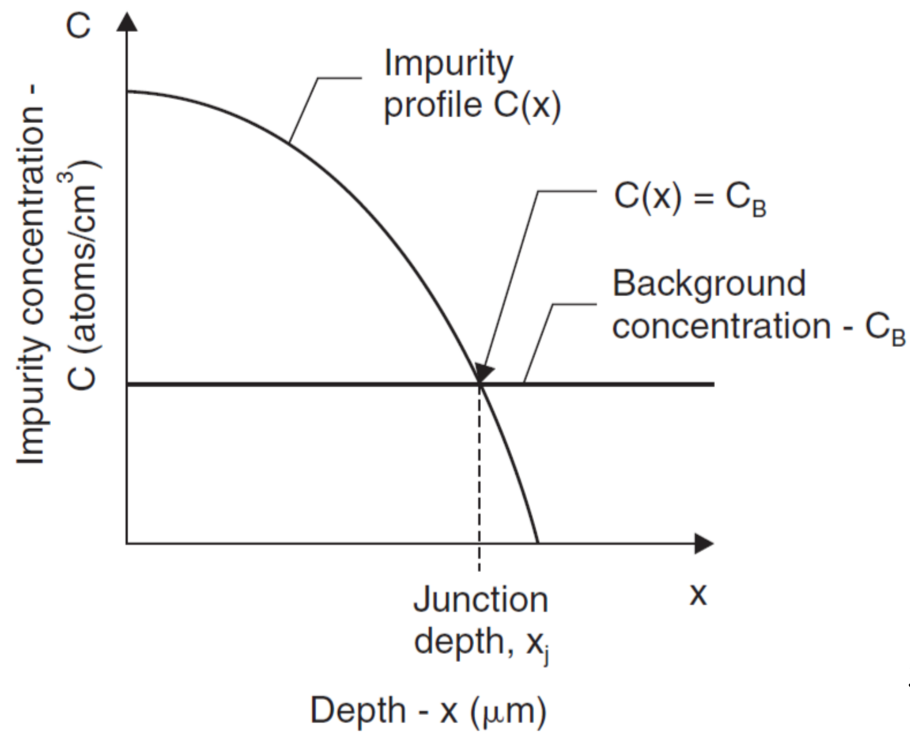
Step 1: introduce a certain dose of dopants

- Pre-deposition
- Ion implantation

Step 2: High temperature annealing

- To drive in the dopants into the desired depth (for diffusion)
- To repair the damage and activate the dopants (for ion implantation)





Mathematic Model

Fick's first law of diffusion

- Diffusion flux J ($cm^{-2}\ sec^{-1}$) is described as:

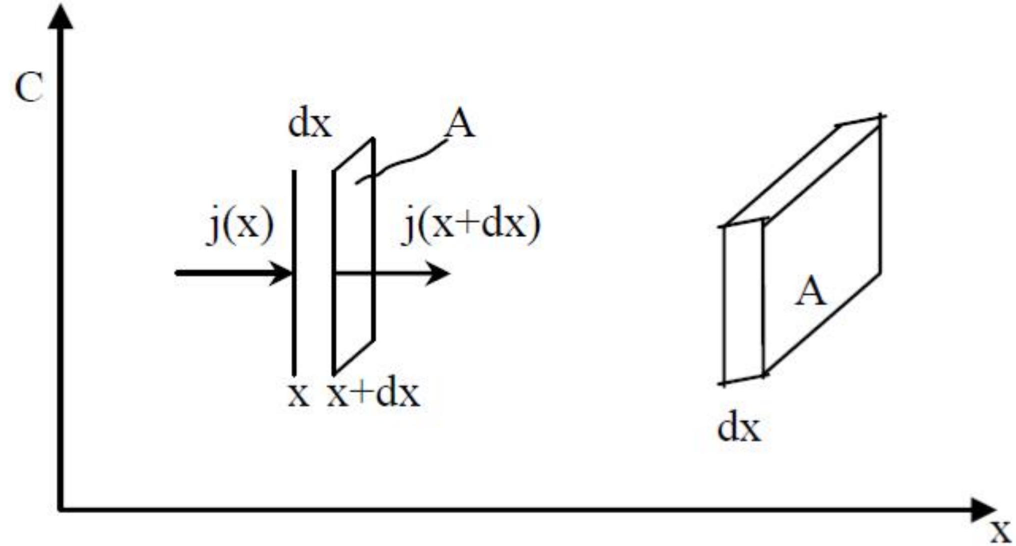
$$\boxed{J = -D\frac{\partial C(x,t)}{\partial x}}$$

Fick's first law of diffusion

- Negative sign "-": the negative sign means that the matter flows in the direction of decreasing concentration.

- *D* : the diffusivity (or diffusion constant, or diffusion coefficient), in a unit of cm^2/sec.

Fick's second law of diffusion



- From the law of conservation of matter: change of concentration with time (dt) = local decrease of the diffusion flux (assume there is no source or sink), we have:

$$dC(x,t) = \frac{[J(x) - J(x+dx)]dtA}{Adx} \qquad \frac{dC(x,t)}{dt} = -\frac{dJ(x,t)}{dx}$$

- Consider $dJ(x)/dx$:

$$\frac{dJ(x)}{dx} = \frac{J(x+dx) - J(x)}{dx}$$

$$\boxed{\frac{\partial C(x,t)}{\partial t} = -\frac{\partial J(x,t)}{\partial x}}$$

- $C(x,t)$: local concentration at position "x".
- $J(x)$: diffusion flux (through a area "A") at position "x".

- Combing with the Fick's first law,

$$\frac{\partial C(x,t)}{\partial t} = \frac{\partial}{\partial x}\left(D \frac{\partial C(x,t)}{\partial x} \right)$$

- When D (in the unit of cm^2/sec) is constant with respect to x:

$$\boxed{\frac{\partial C(x,t)}{\partial t} = D \frac{\partial^2 C(x,t)}{\partial x^2}}$$

Fick's second law of diffusion

- $C(x,t)$ can be solved from the above **partial differential equation** subject to one initial condition and two boundaries conditions.

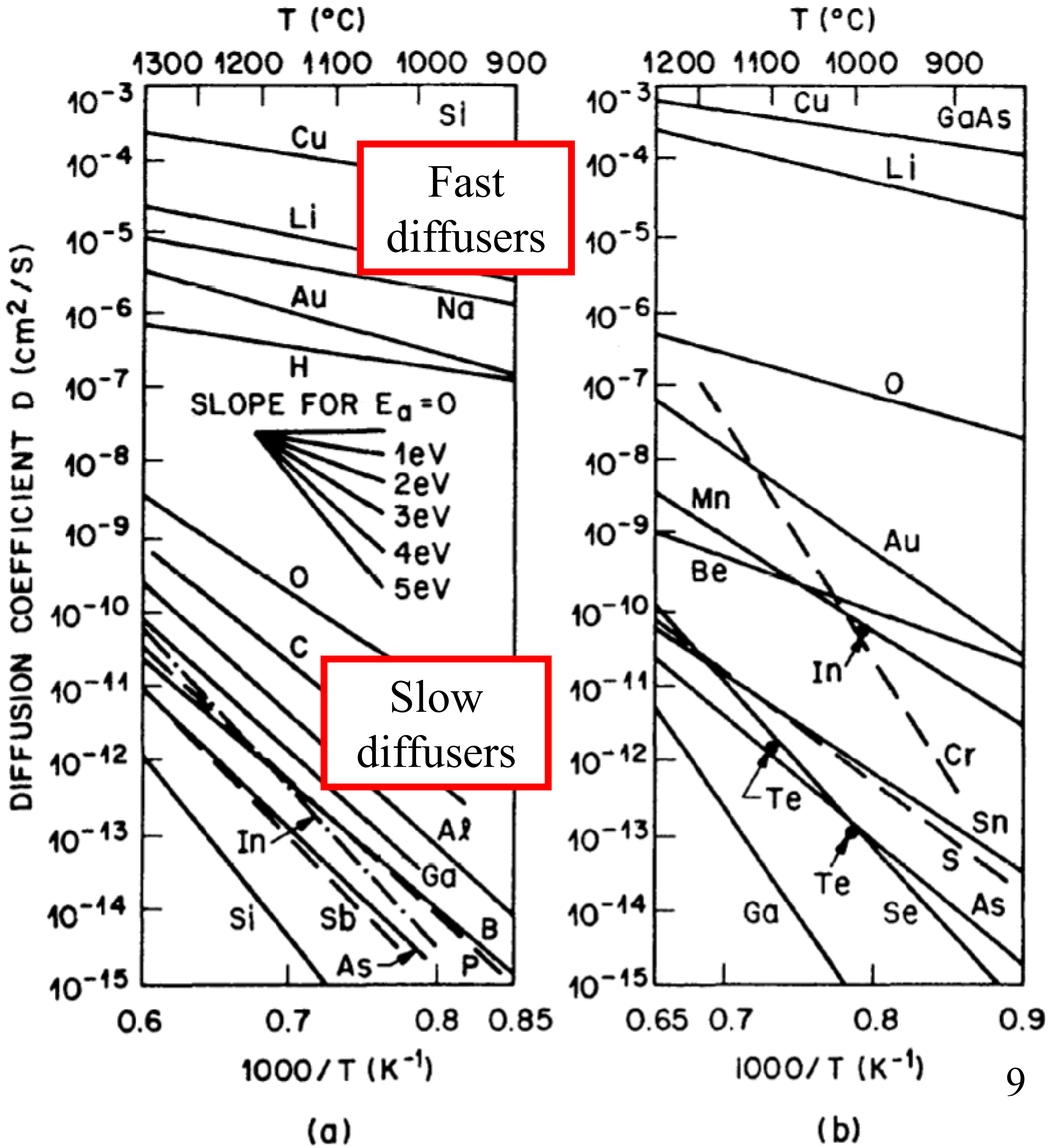
Diffusion Coefficient (*D*)

$$D = D_0 \exp\left(-\frac{E_A}{kT}\right)$$

Two parameters:

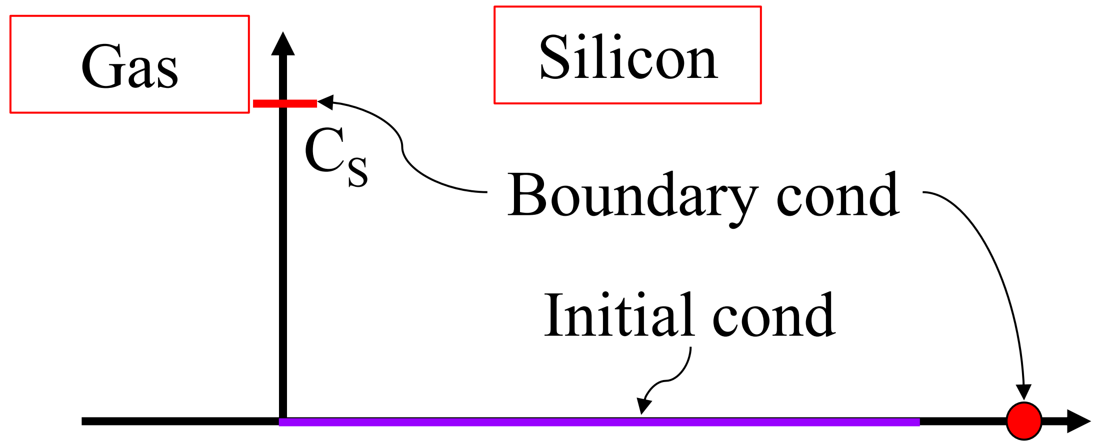
- D_0: denotes the diffusion coefficient extrapolated to infinite temperature
- E_A: stands for the Arrhenius activation energy (eV)

k: Boltzmann's constant
T: absolute temperature



CASE1 : Constant Surface Concentration Diffusion

$$\boxed{\frac{\partial C(x,t)}{\partial t} = D\frac{\partial^2 C(x,t)}{\partial x^2}}$$



- Initial condition

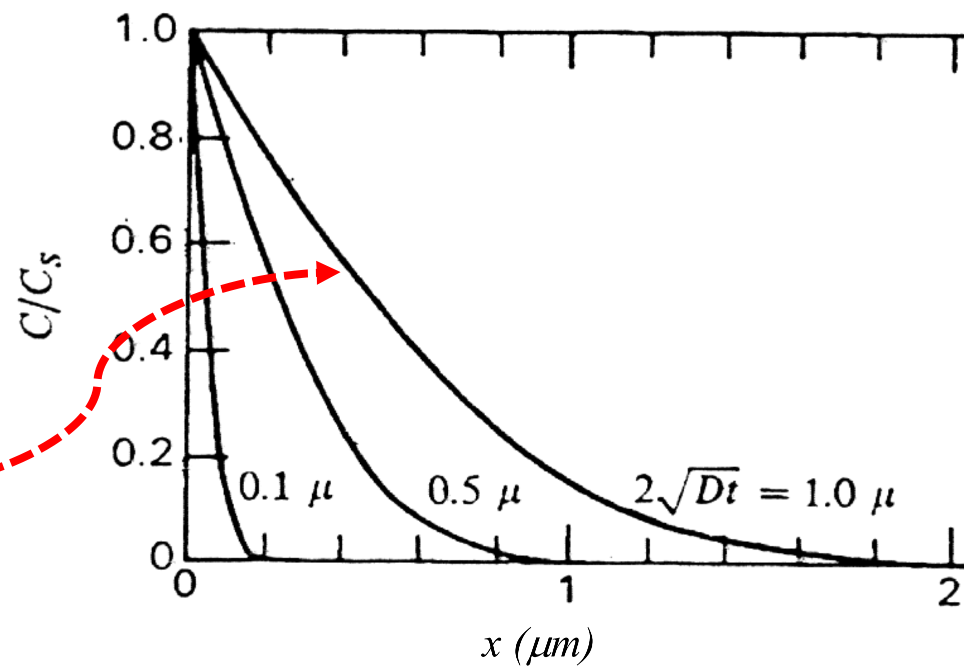
$$C(x,0) = 0 \qquad (t=0)$$

- Boundary conditions ,

$$C(0,t) = C_S \qquad (x=0)$$

$$C(\infty,t) = 0 \qquad (x=+\infty)$$

$$\boxed{C(x,t) = C_S\, erfc(\frac{x}{2\sqrt{Dt}})}$$



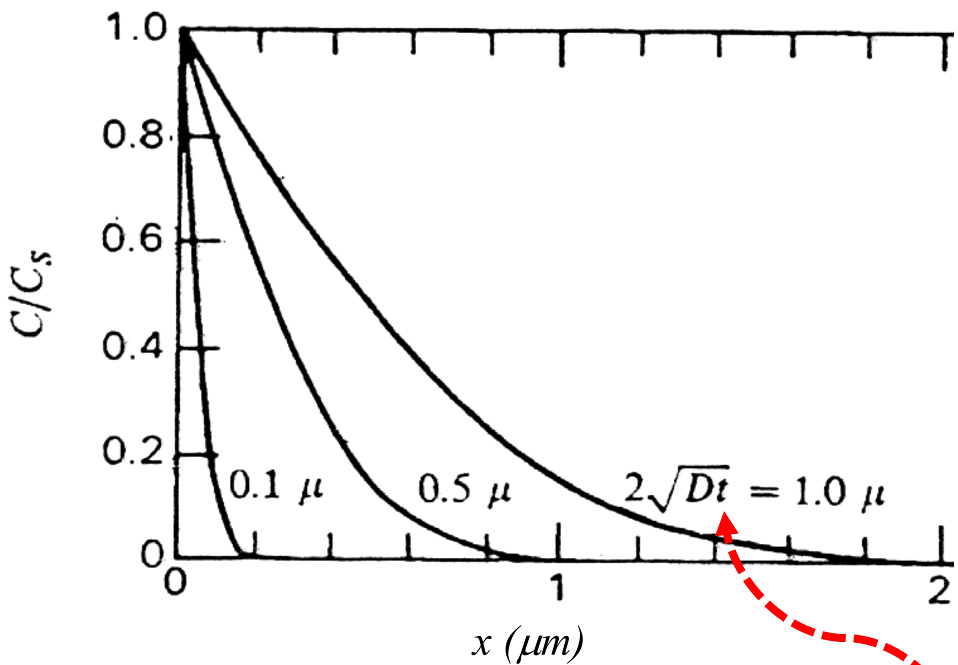
Error function

$$
\begin{aligned}
erf\,(x) &\equiv \frac{2}{\sqrt{\pi}}\int_0^x e^{-y^2}\,dy \\
erfc\,(x) &\equiv 1 - erf(x) \\
erf\,(0) &= 0 \\
erf\,(\infty) &= 1 \\
erf\,(x) &\cong \frac{2}{\sqrt{\pi}}x \qquad for \quad x << 1 \\
erfc\,(x) &\cong \frac{1}{\sqrt{\pi}}\frac{e^{-x^2}}{x} \qquad for \quad x >> 1 \\
\frac{d}{dx}erf\,(x) &= \frac{2}{\sqrt{\pi}}e^{-x^2} \\
\frac{d^2}{dx^2}\,erf\,(x) &= -\frac{4}{\sqrt{\pi}}xe^{-x^2} \\
\int_0^x erfc(y')dy' &= x\;\; erfc(x) \;+\; \frac{1}{\sqrt{\pi}}(1 - e^{-x^2}) \\
\int_0^\infty erfc(x)dx &= \frac{1}{\sqrt{\pi}}
\end{aligned}
$$

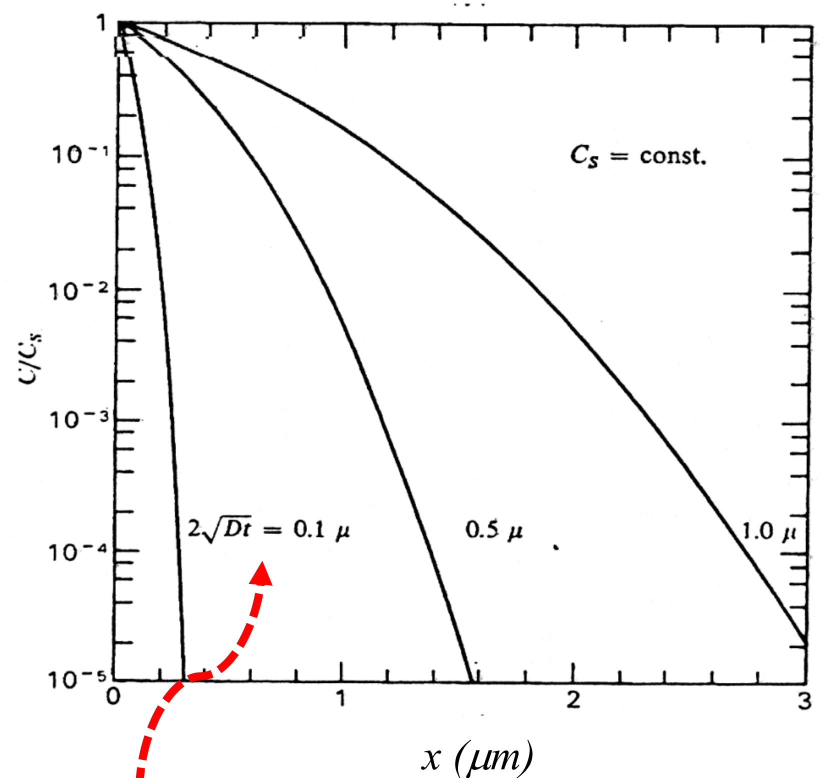
Diffusion profile: *erfc* profile

$$C(x,t) = C_S\, erfc\left(\frac{x}{2\sqrt{Dt}}\right)$$

Linear scale

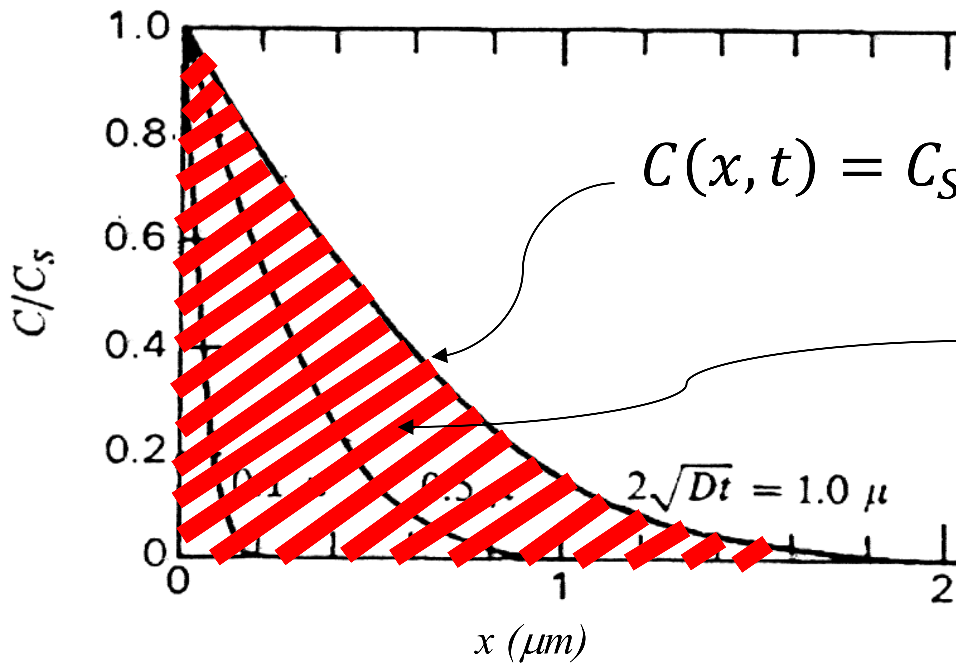


Log scale



$2\sqrt{Dt}$ is defined as diffusion length

Total dopant diffused (dose) : *Q(t)*



$$C(x,t) = C_S\, erfc(\frac{x}{2\sqrt{Dt}})$$

Q(t): equals to the area of the red shaded region

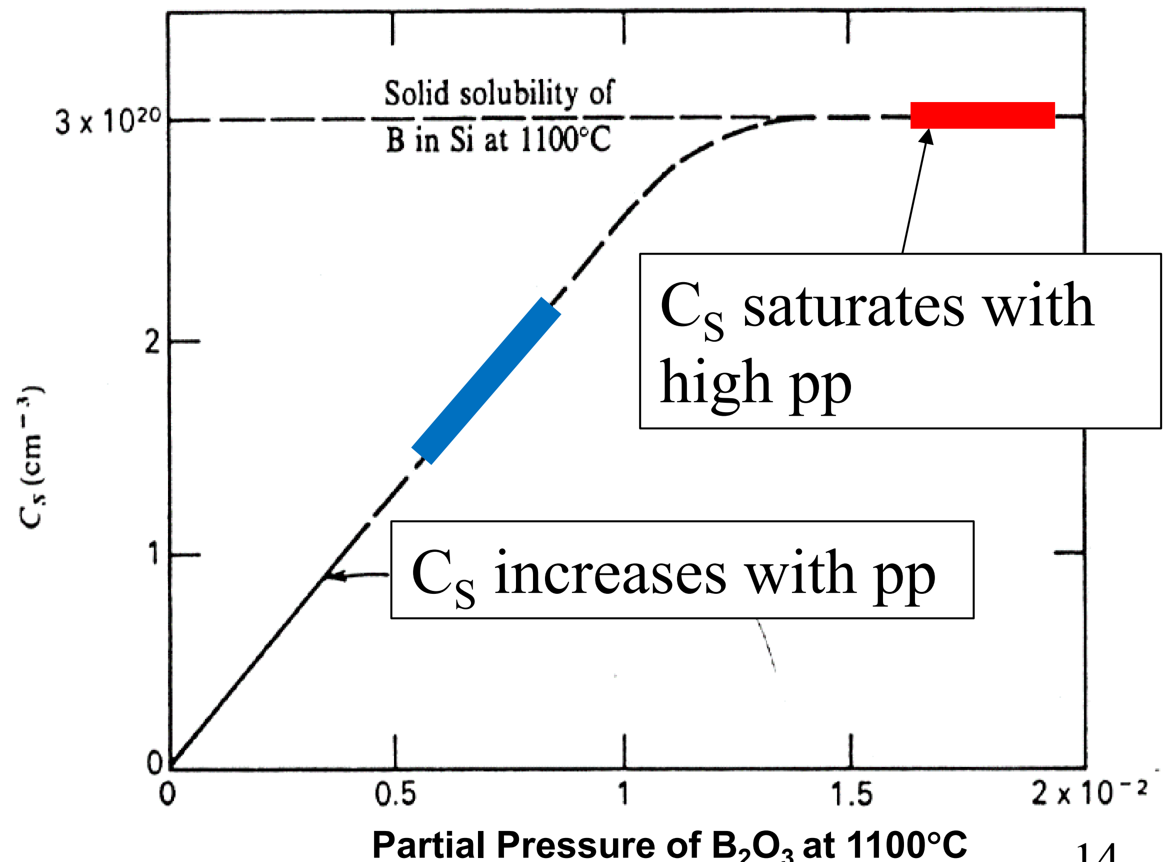
$$Q(t) = \int_0^\infty C(x,t)dx = \int_0^\infty C_S \cdot \mathrm{erfc}\left(\frac{x}{2\sqrt{Dt}}\right) dx$$

$$\boxed{Q(t) = \frac{2C_S\sqrt{Dt}}{\sqrt{\pi}}}$$

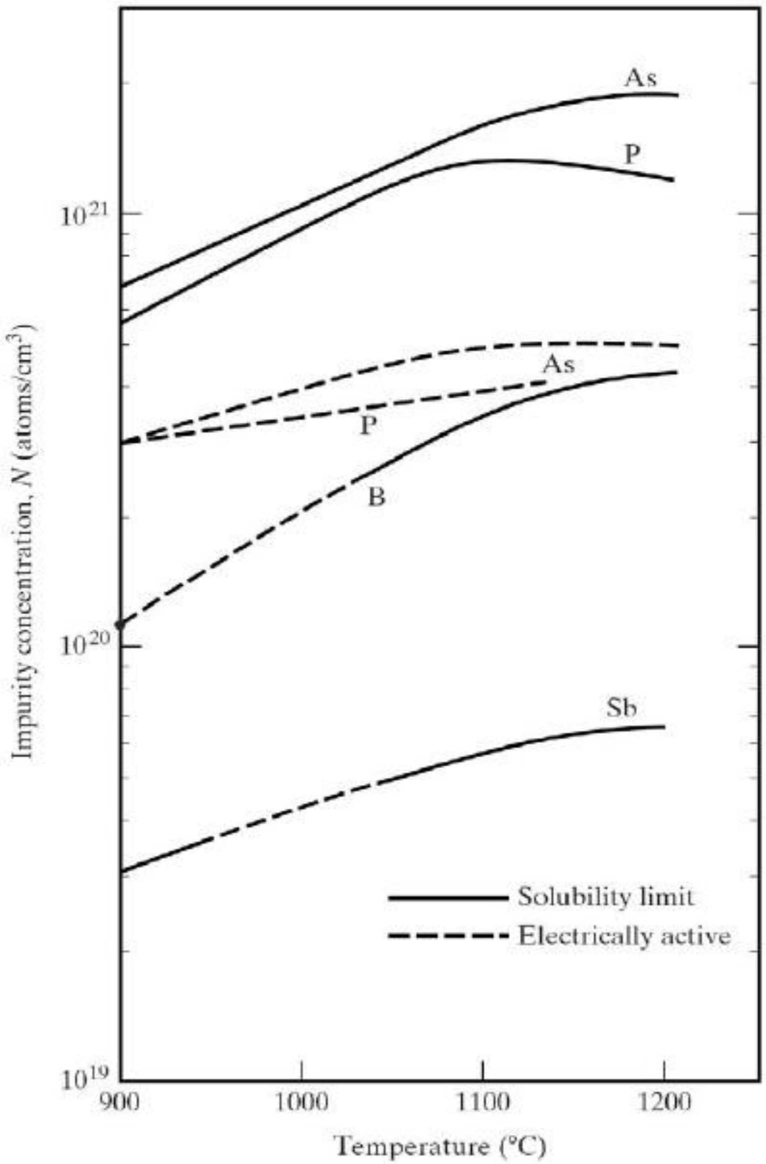
How to achieve C_S constant?

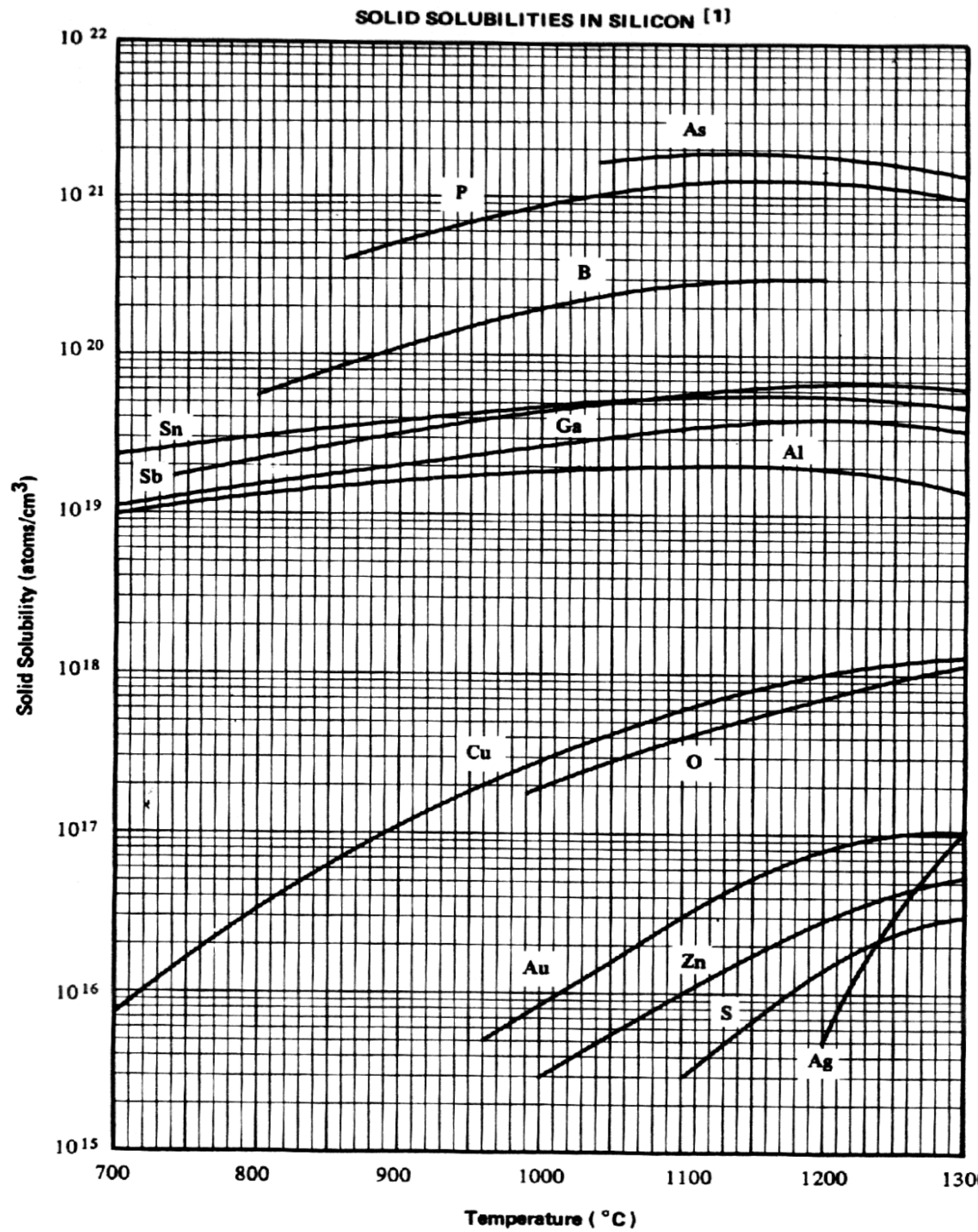
Solid solubility : A maximum concentration of any impurity that can be accommodated in a solid at any given temperature.

- C_S (Si surface) relates to the partial pressure (p.p.) of dopants in the gas
- Diffusion under Solid Solubility Limit
  - Good control of C_S.
  - C_S only depends on process temperature (via solid solubility).
  - Diffusion profile depends on temperature and time



Solid solubility of dopants in Si





CASE 2 : Constant Total Dopant Diffusion

$$\boxed{\frac{\partial C(x,t)}{\partial t} = D\frac{\partial^2 C(x,t)}{\partial x^2}}$$

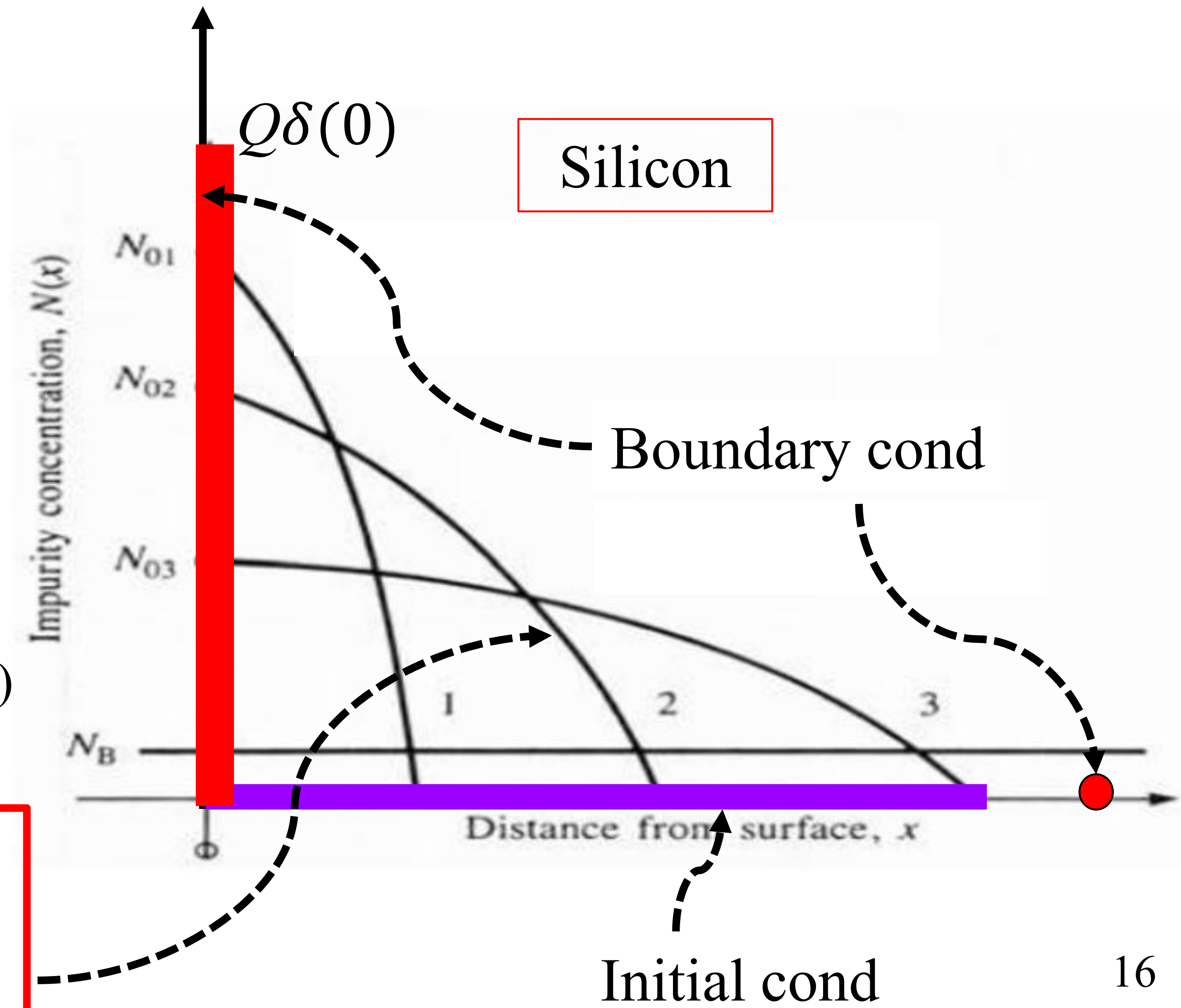
- Initial condition

$$C(x,0)=0 \qquad (t=0)$$

- Boundary condition

$$C(\infty,t)=0$$

$$\int_0^\infty C(x,t)dx = Q \quad (Q\text{=constant})$$

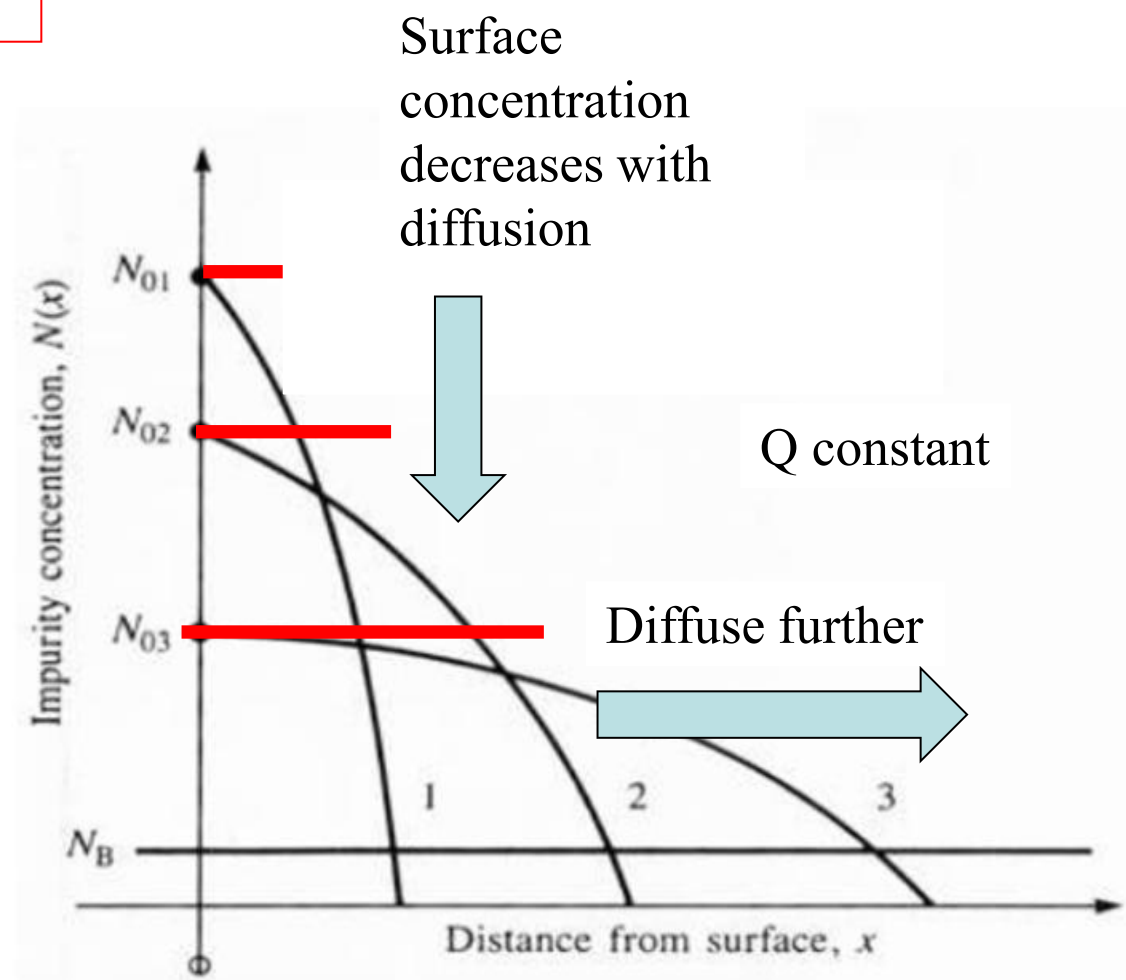


$$\boxed{\begin{aligned} C(x,t) &= \frac{Q}{\sqrt{\pi D t}}\exp(-\frac{x^2}{4Dt}) \\ &= C_S\exp(-\frac{x^2}{4Dt}) \end{aligned}}$$

Diffusion profile: Gaussian profile

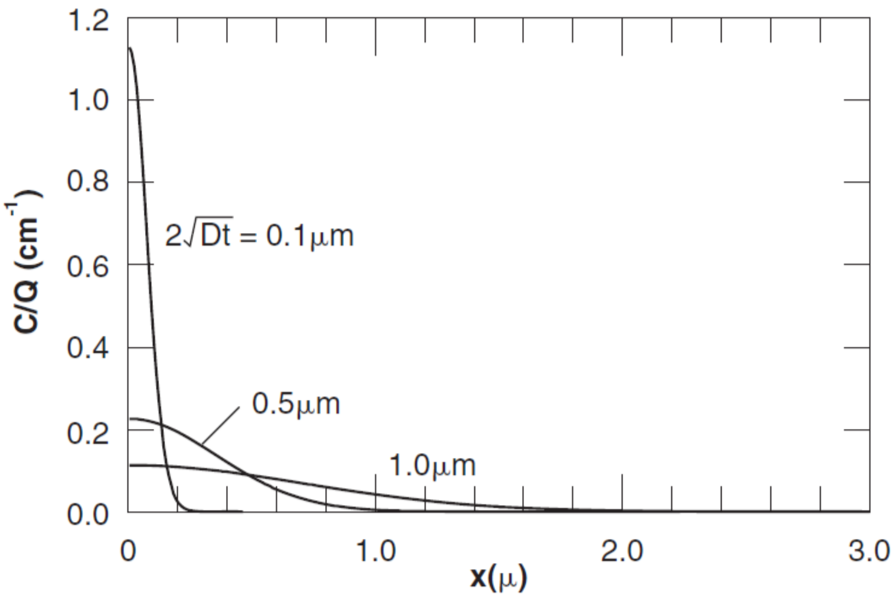
$$C(x,t) = \frac{Q}{\sqrt{\pi D t}}\exp\left(-\frac{x^2}{4Dt}\right)$$
$$= C_S \exp\left(-\frac{x^2}{4Dt}\right)$$

- Total dose of dopant (Q) remain unchanged. (i.e., area under diffusion profile)
- Concentration at Si surface decreases with further diffusion.

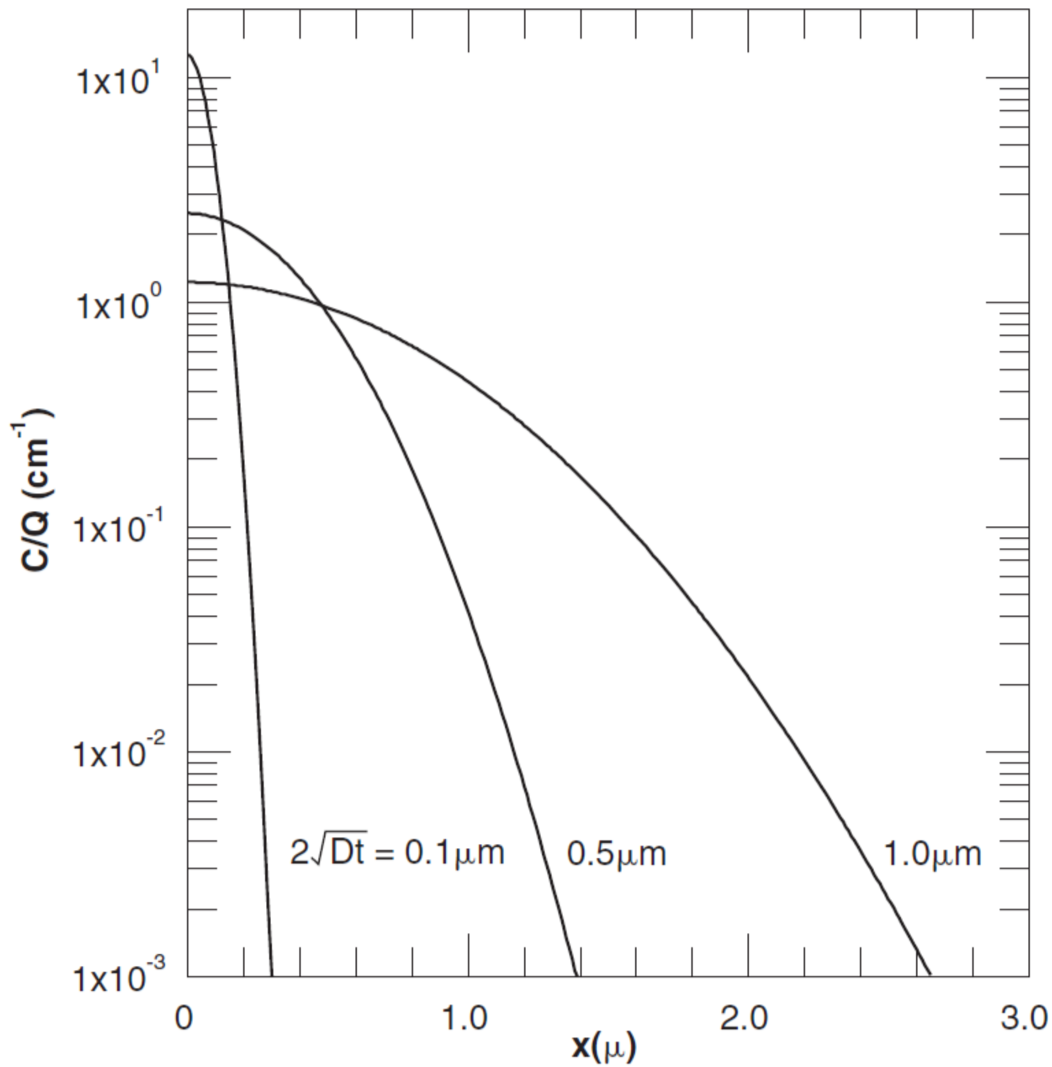


Diffusion profile: linear and log plots

Linear scale



Log scale



Diffusion profile comparison

Complementary error function profile

vs

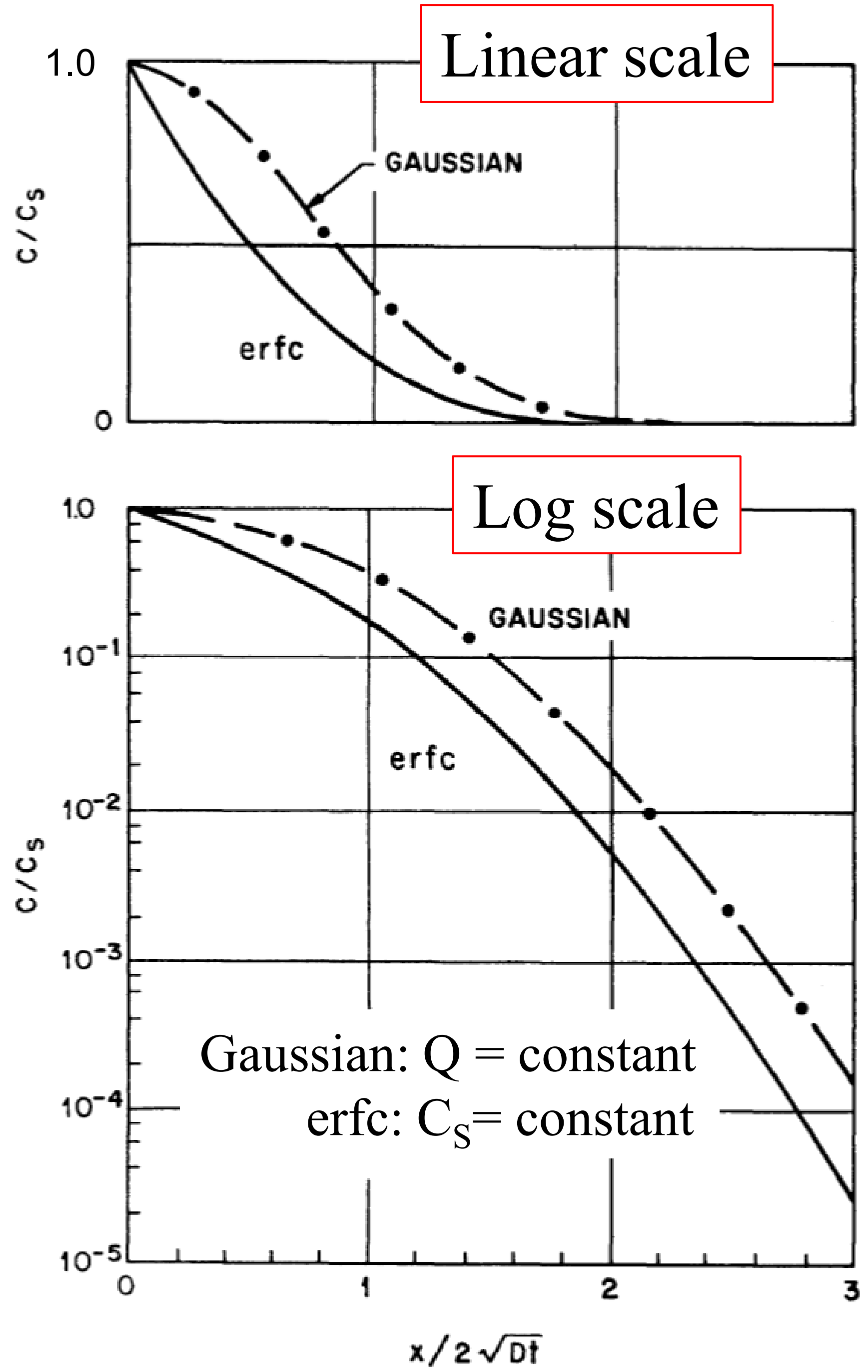
Gaussian function profile

$$erf\,(x) \equiv \frac{2}{\sqrt{\pi}}\int_0^x e^{-y^2}\,dy$$

$$erfc\,(x) \equiv 1 - erf(x)$$

$$erf\,(0) = 0$$

$$erf\,(\infty) = 1$$



Junction Depth

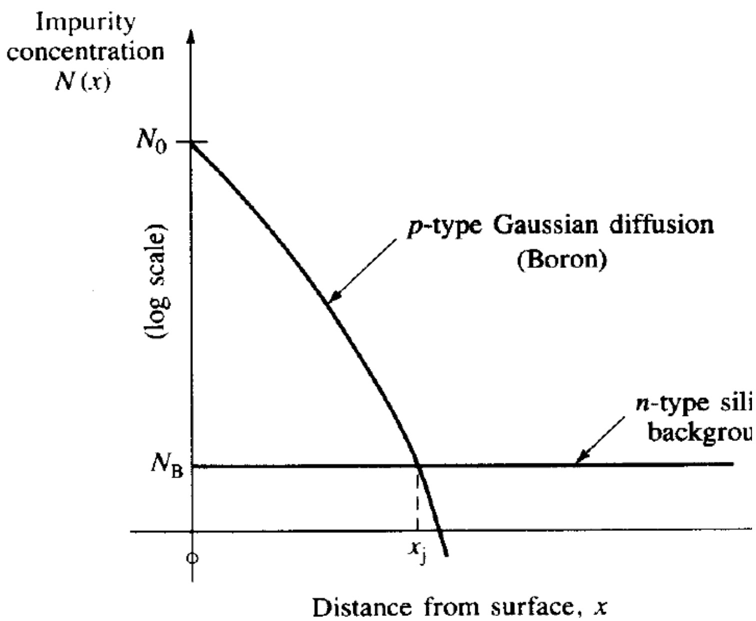
x_j = point at which diffused impurity profile intersects the background concentration, C_B

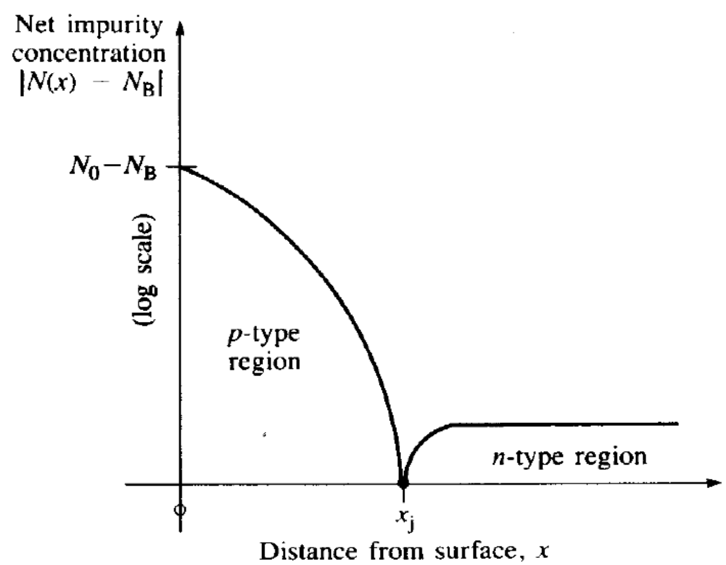
For a complementary error function profile:

$$C(x_j, t) = C_S \operatorname{erfc}\left(\frac{x_j}{2\sqrt{Dt}}\right) = C_B \Longrightarrow x_j = 2\sqrt{Dt}\, erfc^{-1}\left(\frac{C_B}{C_S}\right)$$

For Gaussian dopant profile:

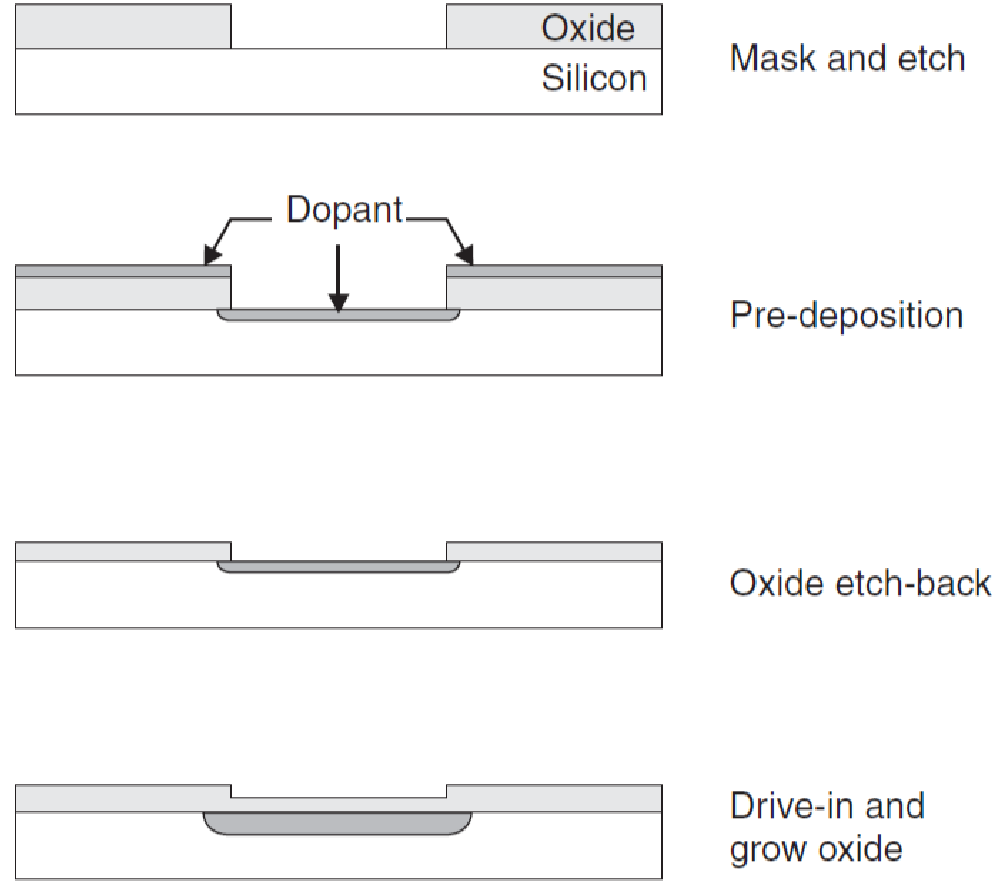
$$C(x_j, t) = C_S \exp\left(-\frac{x_j^2}{4Dt}\right) = C_B \Longrightarrow x_j = 2\sqrt{Dt \ln\left(\frac{C_S}{C_B}\right)}$$





Typical Two-Step Diffusion

- Typical two step diffusion procedure:
  - Step 1: pre-deposition (i.e., constant surface concentration diffusion)
  - Step 2: drive-in diffusion (i.e., constant total dopant diffusion)
- Oxide used as mask for selective diffusion
- Oxide etch-back is to prevent dopants in mask oxide to diffuse during drive-in step.
- Drive-in step is normally performed in oxidizing ambient and oxide grown prevents dopant out-diffusion



Two-step diffusion profile

For processes where there are both pre-deposition and drive-in diffusion, the final profile type (i.e., complementary error function or Gaussian) is determined by the one which has a much greater thermal budget (Dt product)

- When $(Dt)_{predep} >> (Dt)_{drive\text{-}in}$, impurity profile is complementary error function
- When $(Dt)_{drive\text{-}in} >> (Dt)_{predep}$, impurity profile is Gaussian function (which is usually the case)

Successive Diffusions

- A wafer typically goes through many time-temperature cycles including pre-deposition, drive-in, oxidation, CVD, etc.

- It is of high interest to estimate the final impurity distribution after all these thermal steps.

- These steps take place at different temperatures for different lengths of time.

- The effect of these steps is determined by calculating the total Dt product $(Dt)_{tot}$ for the diffusion.

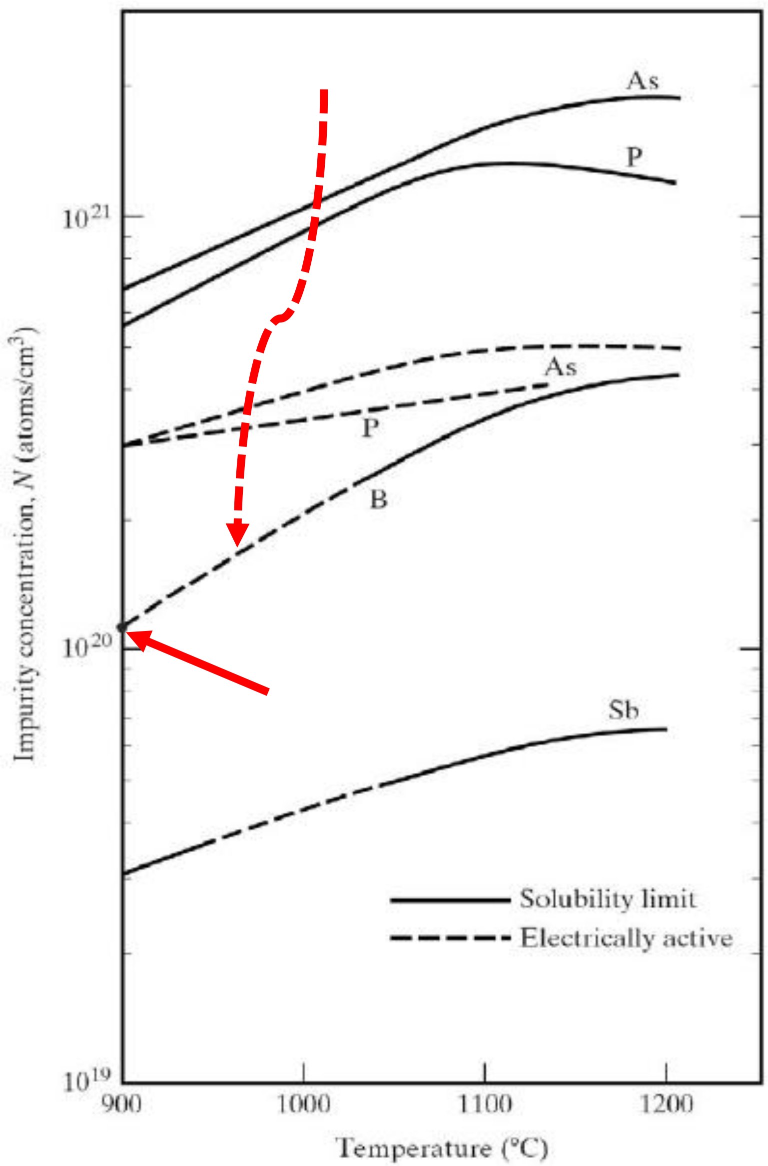
$$(Dt)_{tot} = \sum_i D_i t_i$$

Example: two-step diffusion

A boron diffusion is performed to form the base region of an npn transistor in a 0.18 Ω-cm (i.e. N_D=3x10^{16} /cm^3) n-type Si wafer. A solid-solubility-limited boron pre-deposition is performed at 900 ^oC for 15 min followed by a 5 hr drive-in at 1100 ^oC. Assume D_O=10.5 cm^2/sec and E_A=3.69 eV of B diffusion in Si.

Find the B surface concentration and junction depth (a) after the pre-deposition step, and (b) after the drive-in step.

(a) Find B surface concentration and junction depth after the pre-deposition step.



Solution:

(a) Pre-deposition, T_1=900°C, t_1=15 mins

T_1=900 °C=1173 K

From the Fig in the left, C_{S1}=1.1E20 cm^{-3}

$$D_1 = D_0 exp(-\frac{E_A}{kT_1})$$

$$D_1 = 10.5\exp(-\frac{3.69eV}{(8.614 \times 10^{-5} eV/K) \times 1173K}$$

$$= 1.45 \times 10^{-15} \text{ cm}^2\text{/sec}$$

$$t_1 = 15 \min = 900\ sec$$

$$D_1 t_1 = 1.31 \times 10^{-12}\ cm^2$$

erfc diffusion profile

$C_B = N_D$

$$C_1(x_{j1},\, t_1) = C_{s1}\,\mathrm{erfc}\left(\frac{x_{j1}}{2\sqrt{D_1 t_1}}\right) = \mathrm{C_B}$$

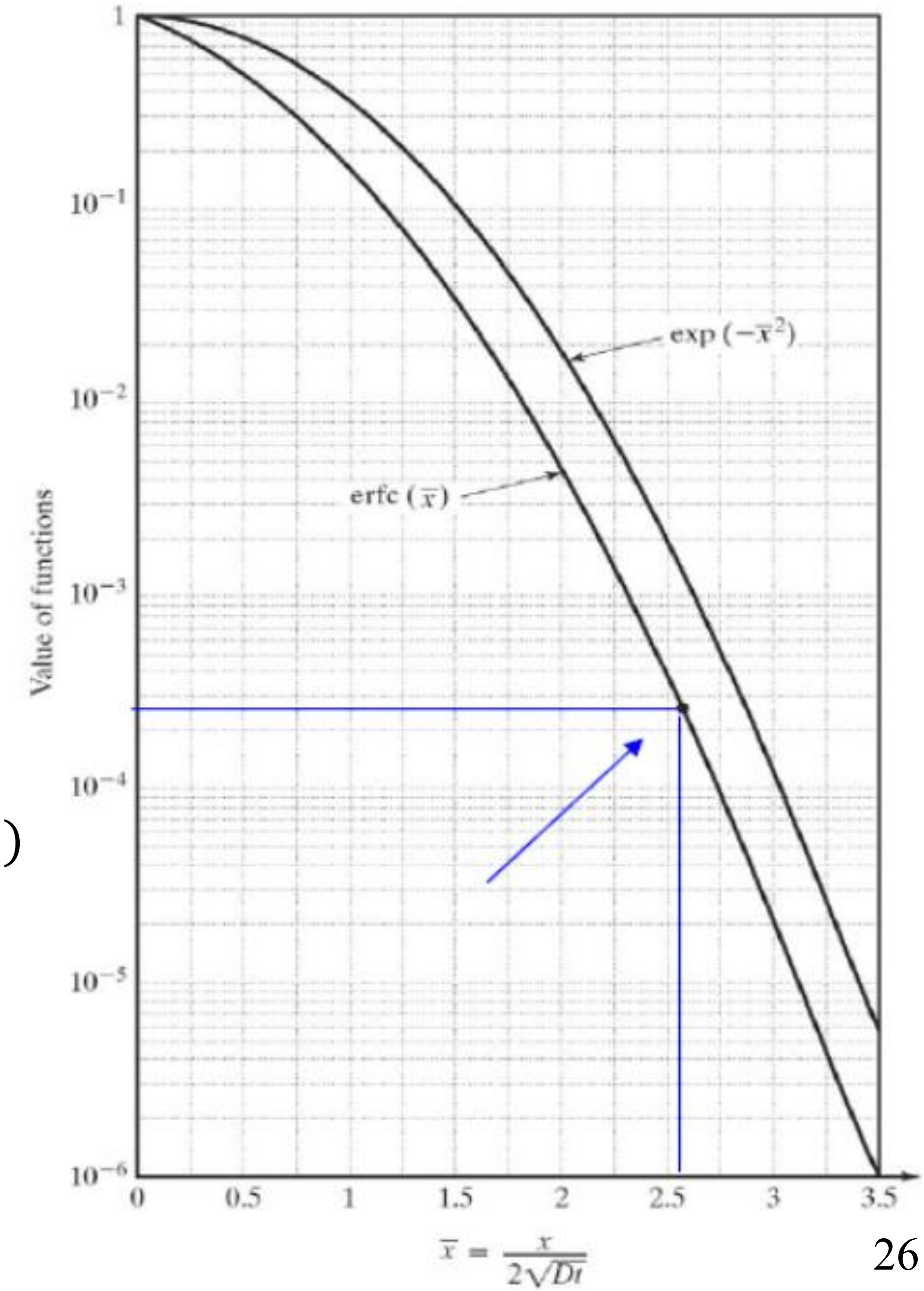
$$x_{j1} = 2\sqrt{D_1 t_1}\, erfc^{-1}\left(\frac{C_B}{C_{S1}}\right)$$

$$x_{j1} = 2 \times \sqrt{1.31 \times 10^{-12}}\, erfc^{-1}\left(\frac{3\times10^{16}}{1.1\times10^{20}}\right)$$

$$= 2.28 \times 10^{-6}\ cm\ erfc^{-1}(2.73 \times 10^{-4})$$

$$= 2.28 \times 10^{-6}\ cm\ (2.57)$$

$$= 5.86 \times 10^{-6}\ cm = 0.0586\ \mu m$$



(b) Find B surface concentration and junction depth after the drive-in step.

Solution:

(b) Drive-in, T_2=1100 °C, t_2= 5 hrs

T_2=1100 °C=1373 K

$$t_2 = 5 \text{ hrs} = 18000 \, sec$$

$$D_2 = D_0 exp(-\frac{E_A}{kT_2})$$

$$D_2 = 10.5\exp(-\frac{3.69eV}{(8.614\times10^{-5} eV/K)\times1373K})$$

$$= 2.96 \times 10^{-13} \text{ cm}^2/\text{sec}$$

$$D_2 t_2 = 5.33 \times 10^{-9} \, cm^2$$

$$D_2 t_2 \gg D_1 t_1$$

The profile after pre-deposition can be treated as a delta function, the diffusion profile after drive-in step can be predicted using Gaussian profile with Q=Q_1 introduced during pre-deposition step.

Q_1: the dopant introduced during predeposition step

Gaussian diffusion profile

$$Q_1 = \frac{2C_{S1}\sqrt{D_1 t_1}}{\sqrt{\pi}} = 1.42 \times 10^{14}\ /cm^2$$

$C_B = N_D$

$$C_2(x_{j2},\ t_2) = \frac{Q_1}{\sqrt{\pi D_2 t_2}}\exp\left(-\frac{x_{j2}^2}{4D_2 t_2}\right) = C_B$$

$$C_{S2} = \frac{Q_1}{\sqrt{\pi D_2 t_2}}$$

$$= \frac{1.42 \times 10^{14}}{\sqrt{3.14 \times 5.33 \times 10^{-9}}}$$

$$= 1.1 \times 10^{18}/cm^3$$

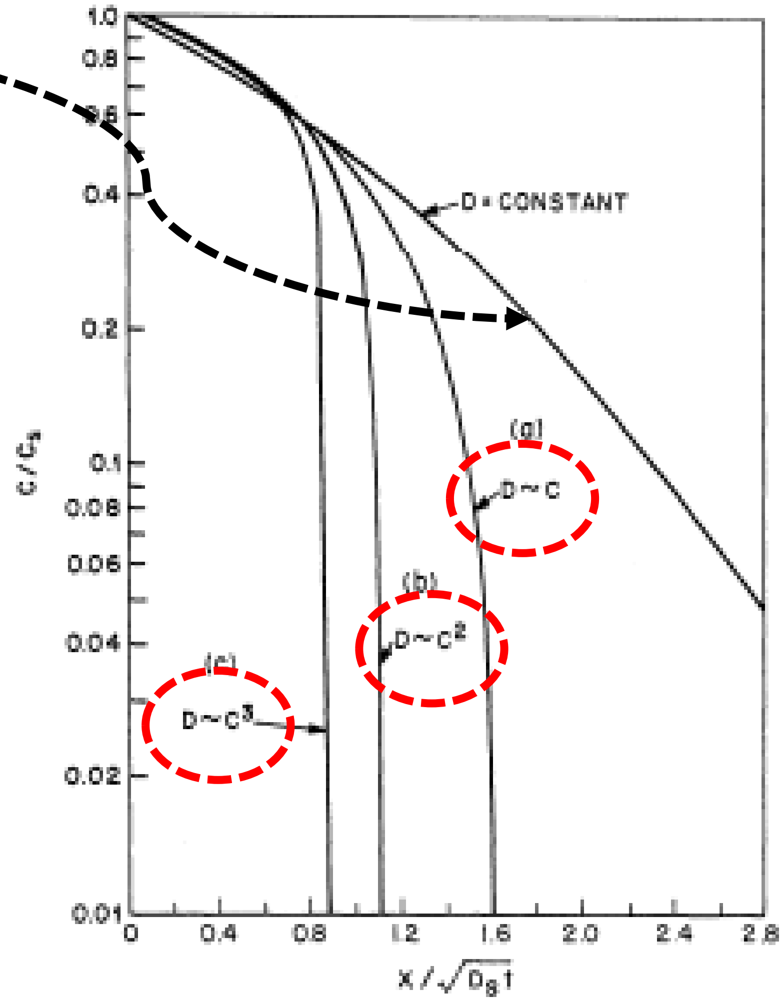
$$x_{j2} = 2\sqrt{D_2 t_2 \ln\left(\frac{C_{S2}}{C_B}\right)}$$

$$= 2\sqrt{5.33 \times 10^{-9} \ln\left(\frac{1.1 \times 10^{18}}{3 \times 10^{16}}\right)}$$

$$= 2.77 \times 10^{-4}\ cm = 2.77\ \mu m$$

Concentration Dependent Diffusion

$$\boxed{\frac{\partial C(x,t)}{\partial t} = D\frac{\partial^2 C(x,t)}{\partial x^2}}$$



- The above equation assumes that "D" is independent of position (x)
- When C(x) is lower than n_i at the diffusion temperature, good approximation is obtained. (e.g. $n_i = 5\times10^{18}$ cm^{-3} at 1000°C)
- When C(x) is higher, significant deviation from the model due to **concentration dependence of diffusivity**.

D is a function of n,
which is a function of x,
so D is a function of x.

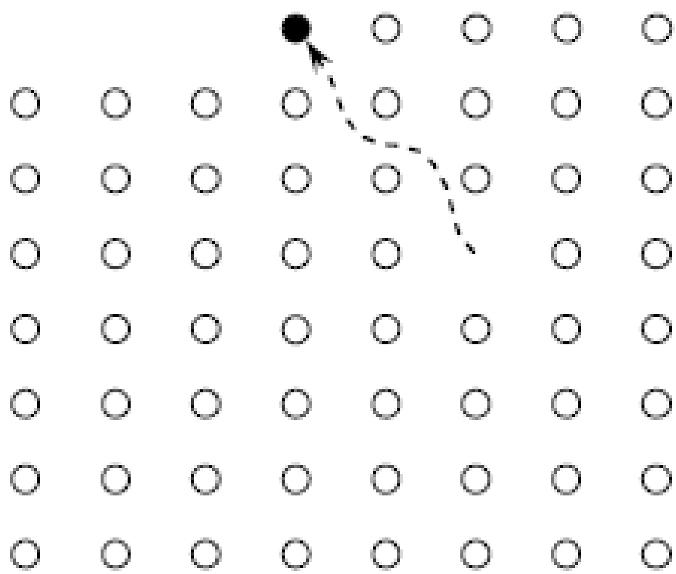
$$\frac{\partial C(x,t)}{\partial t} = \frac{\partial}{\partial x}\left(D \frac{\partial C(x,t)}{\partial x} \right) = \frac{\partial D}{\partial x}\frac{\partial C}{\partial x} + D\frac{\partial^2 C}{\partial x^2}$$

- The above partial differential equation is hard to solve in a closed form.
- Numerical calculation technique is required.
- "Atomistic diffusion model" is developed to model "D" at high concentration.

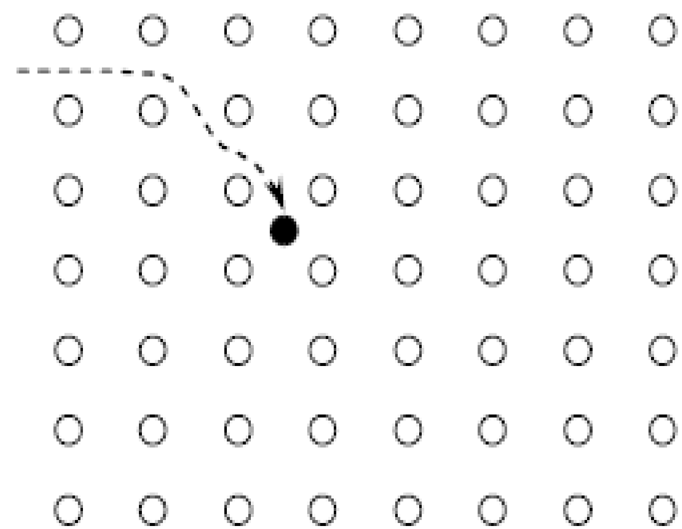
Atomistic Models of Diffusion

- Point defects
- Dopant diffusion mechanisms
- Fair-Tsai vacancy model

Point Defects



Vacancy defect

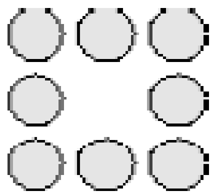


Interstitial defect

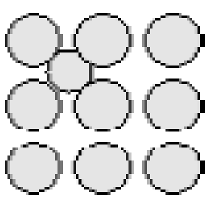
Intrinsic and extrinsic point defects

Intrinsic defects

- Vacancy (denoted V): an atom is removed.

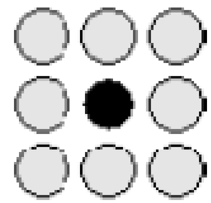


- Self-Interstitial (denoted I): a host atom sits in a normally unoccupied site or interstice .

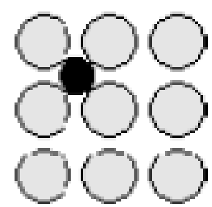


Extrinsic defects
(due to an impurity)

- Substitutional defect, such as carbon substitutional defect (denoted C_s)



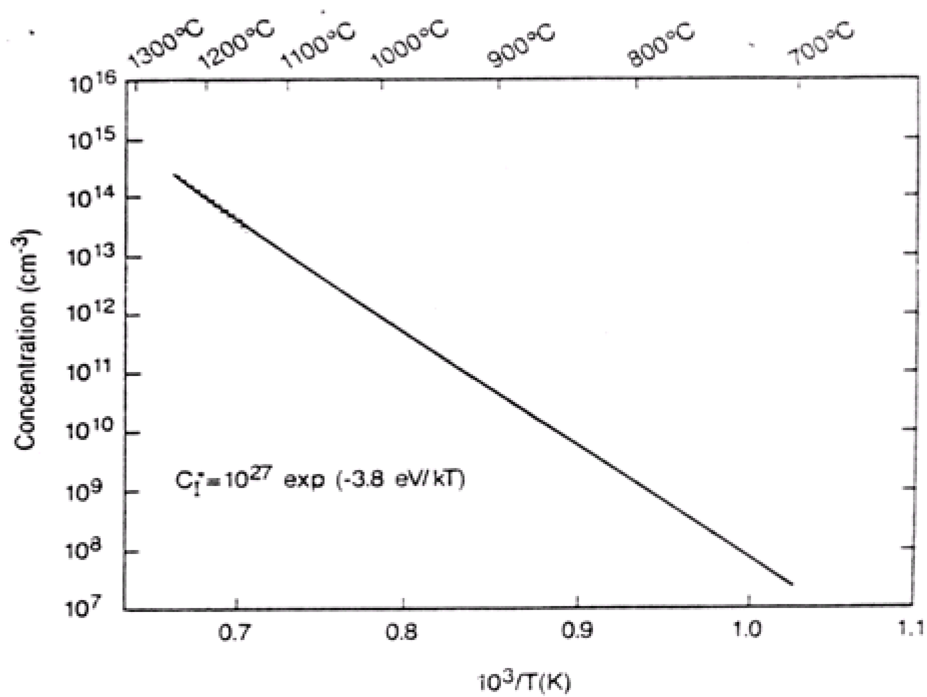
- Interstitial (such as the carbon interstitial (denoted C_i).



Point defects in Si

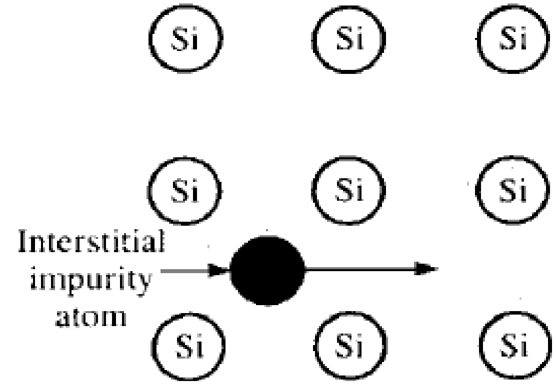
- The point defects play a decisive role in dopant diffusion in silicon.
- At elevated temperatures, lattice atoms vibrate around the lattice sites. Occasionally a host atom acquires sufficient energy to leave the lattice site, becoming **an interstitial atom and creating a vacancy.**
- Thus, point defects exist even in the pure silicon lattice in the form of interstitial and vacancy, and these are called ***native point defects***.

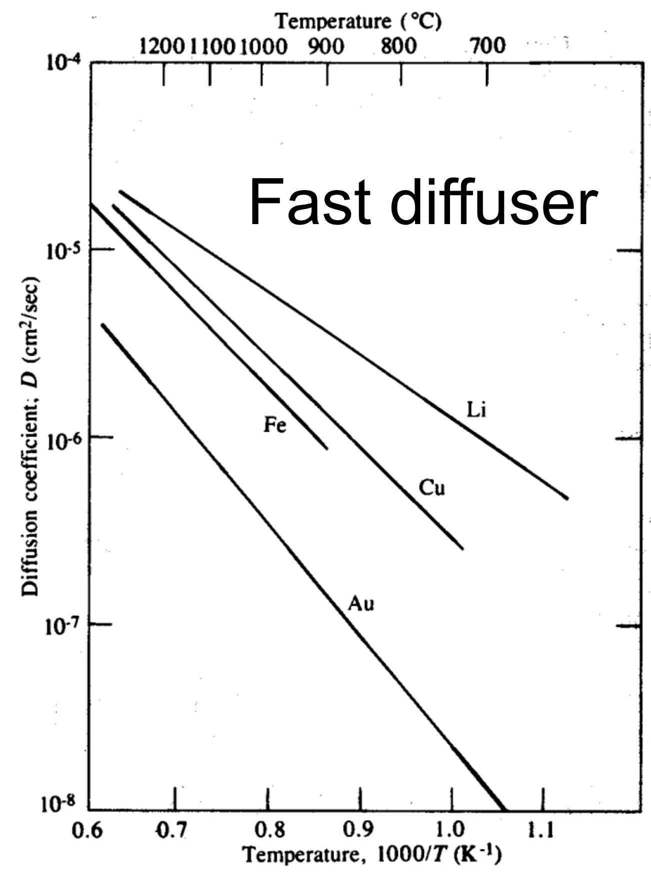
The equilibrium concentration of silicon self-interstitials



Dopant diffusion: Interstitial mechanism

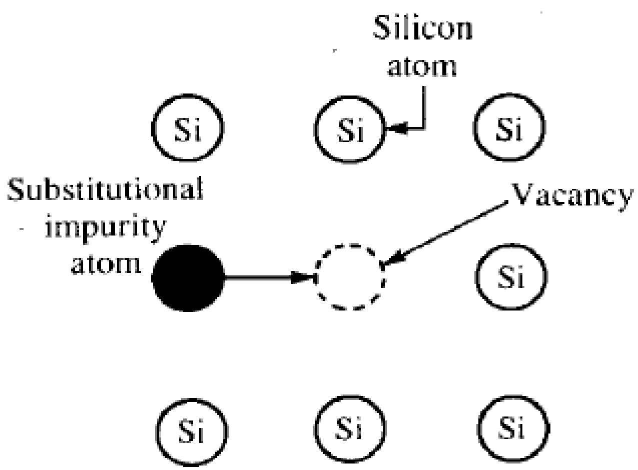
- When an atom moves from one interstitial position to another without occupying a lattice site.
- An atom that is smaller than the host atom and does not form covalent bonds with silicon often moves interstitially.
- The activation energy for this diffusion mechanism is low since no energy is required to form a vacancy.
- Rapid diffusion, hard to control.
- Impurity not in lattice so not electrically active.



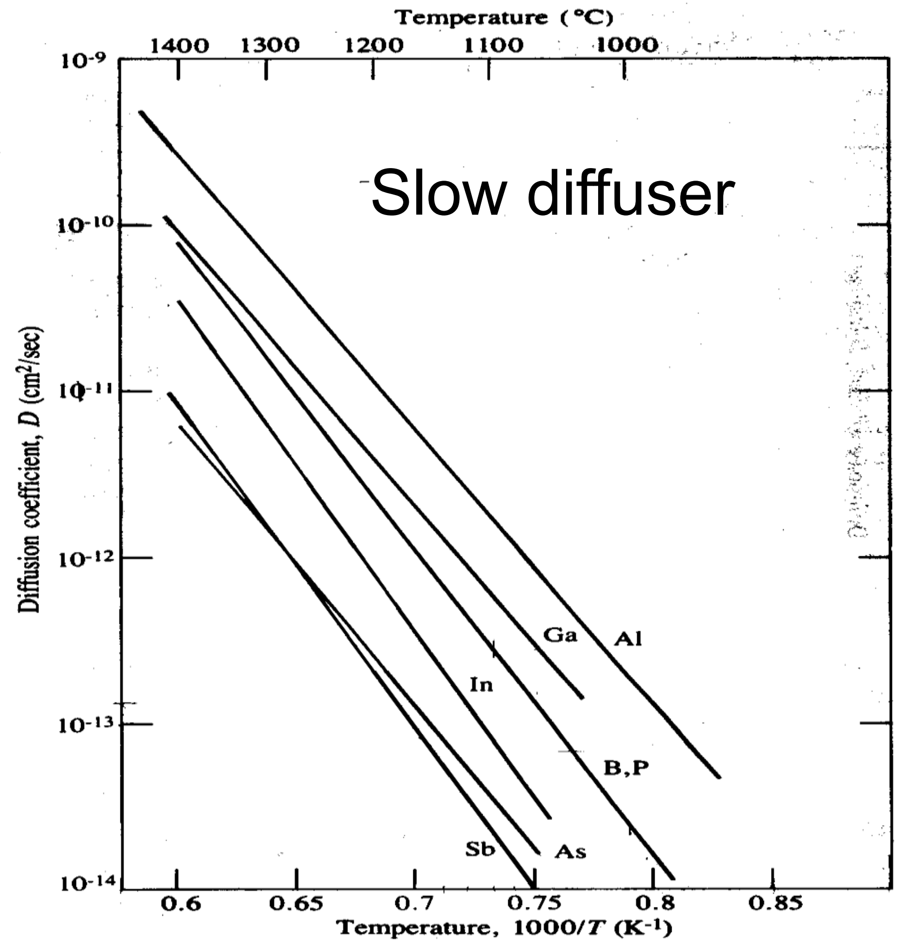


No requirement for Si vacancy or Si interstitial!

Dopant diffusion: Substitutional (vacancy) mechanism

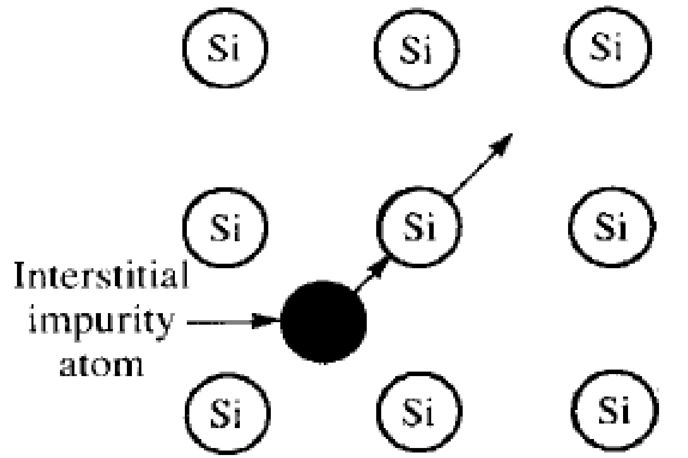


Involve Si vacancy!



- An atom (impurity or Si atom) moves along vacancies in the lattice
- Self-diffusion: when the migrate atom is a host atom (e.g. Si atom)
- Impurity diffusion: when the migrate atom is an impurity atom.

Dopant diffusion: Interstitialcy mechanism

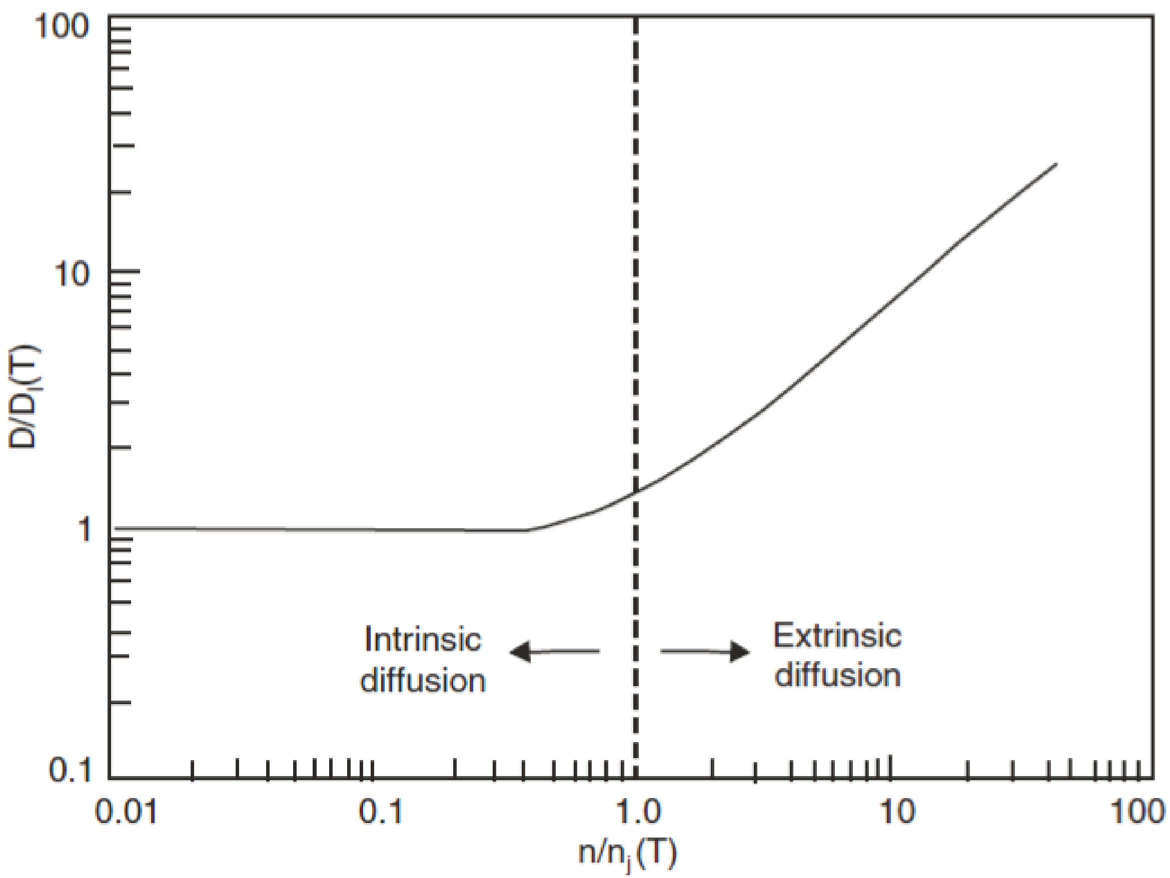


Involve Si interstitial!

- An impurity atom replaces a Si atom in the lattice, and Si atom is displaced to an interstitial site.
- Self-interstitial Si atom displaces an impurity atom, which in turn becomes an interstitial atom.
- Dopant diffusion depends on the concentration of Si interstitial.

Fair-Tsai Vacancy Model

- Intrinsic diffusivity D_i : when $C < n_i$
- Extrinsic diffusivity D_e : when $C > n_i$
- F-T vacancy model (point defect model) is used in SUPREM software simulator.



Intrinsic diffusivity

- F-T model assumes the diffusivity of an ionized dopant atom is based on the sum of the diffusivities of dopant with the help of neutral vacancies and ionized (charged) vacancies
- A single vacancy may exist at least in four charged states: V^+, V^0, V^-, and V^{2-}.

Intrinsic diffusivity

$$D_i = D^0 + D^- + D^{2-} + D^+ + \cdots$$

Neutral vacancy/dopant diffusion

Singly negative charged vacancy/dopant diffusion

Doubly negative charged vacancy/dopant diffusion

Singly positive charged vacancy/dopant diffusion

Extrinsic diffusivity

- During extrinsic diffusion, each contribution must be modified by the ratio of the doping level to the intrinsic carrier concentration raised to the power of the charge state.
- Extrinsic diffusivity (D_e) is a function of dopant concentration.

$$D_e = D^0 + D^- \left(\frac{n}{n_i}\right) + D^{2-}\left(\frac{n}{n_i}\right)^2 + D^+\left(\frac{p}{n_i}\right)$$

Contribution of neutral vacancy

Contribution of singly negative charged vacancy

Contribution of doubly negative charged vacancy

Contribution of singly positive charged vacancy

Diffusion of Major Dopants

- Major dopants

  - Boron : the only p-type dopant used in VLSI
  - Arsenic : shallow junction n+ source/drain
  - Phosphorus : n-well, n- region for LDD NMOS, n+ gate polysilicon

  - Diffusivity is described as

$$D = D_0 \exp(-\frac{E_A}{kT})$$

where E_A is activation energy, D_0 is a pre-factor.
E_A is between 0.6 and 1.2 eV for interstitial diffuser, and between 3 and 4 eV for substitutional diffuser.

Boron

- Intrinsic diffusivity (low concentration)

$$D_B^i = D_B^0 + D_B^+$$

- Extrinsic diffusivity (high concentration)

$$D_B^e = D_B^0 + D_B^+ \left(\frac{p}{n_i}\right)$$

$$D_B^0 = 0.037\exp\left(-\frac{3.46\ eV}{kT}\right)\ \mathrm{cm^2/sec}$$

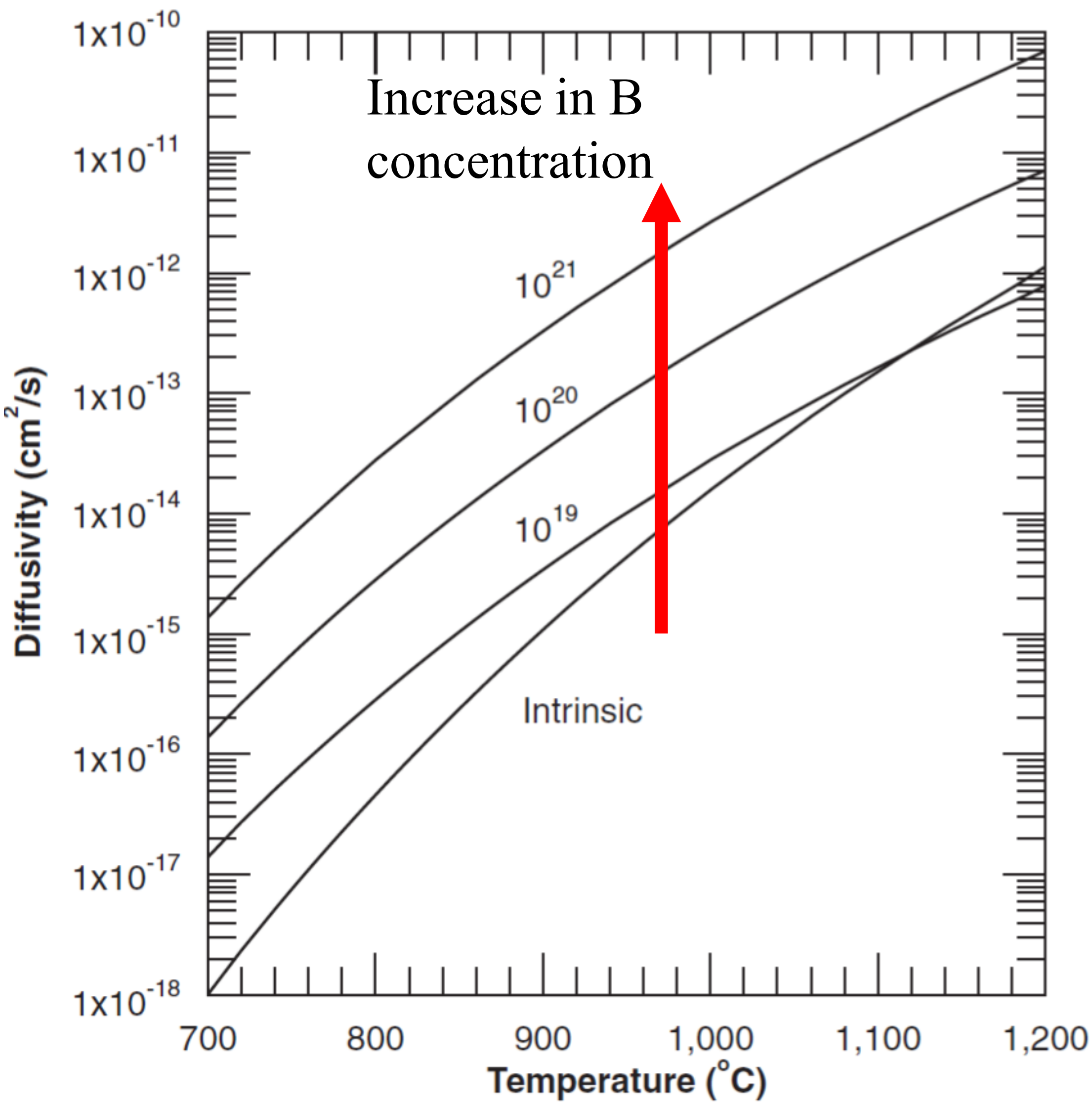
$$D_B^+ = 0.72\exp\left(-\frac{3.46\ eV}{kT}\right)\ \mathrm{cm^2/sec}$$

$$D_B^i = 0.76\exp\left(-\frac{3.46\ eV}{kT}\right)\ \mathrm{cm^2/sec}$$

$$D_B^e \approx D_B^+ \left(\frac{p}{n_i}\right)\ \mathrm{cm^2/sec}$$

- Boron diffuse mainly through BV^O (D_B^0) and BV^+ (D_B^+) pairs.
- B diffusion in Si is dominated in BV^+ pair (single positive charged **point-defect**).
- E_A is lowest among the major dopants. The relatively lower activation energy and less abrupt junction make it difficult to achieve very shallow junction.

Calculated diffusivity of boron in Si



Arsenic

- Intrinsic diffusivity (low concentration)

$$D_{As}^{i} = D_{As}^{0} + D_{As}^{-}$$

- Extrinsic diffusivity (high concentration)

$$D_{As}^{e} = D_{As}^{0} + D_{As}^{-} \left(\frac{n}{n_i}\right)$$

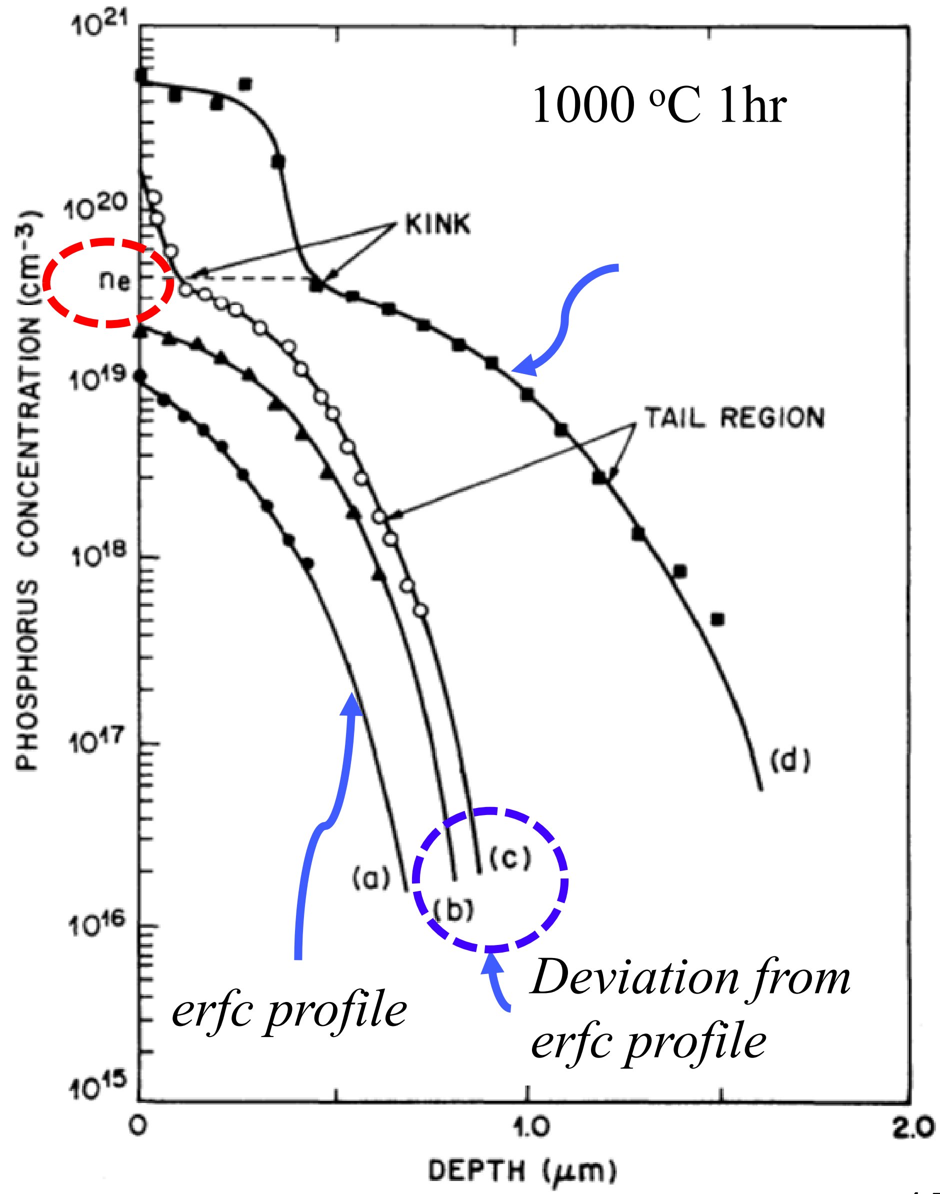
$$D_{As}^{0} = 0.066\exp\left(-\frac{3.44\ eV}{kT}\right)\ \mathrm{cm^2/sec}$$

$$D_{As}^{-} = 12.0\exp\left(-\frac{4.05\ eV}{kT}\right)\ \mathrm{cm^2/sec}$$

- As diffuse mainly through neutral and single negative charged vacancies.
- E_A is the highest among the major dopants.

Phosphorus

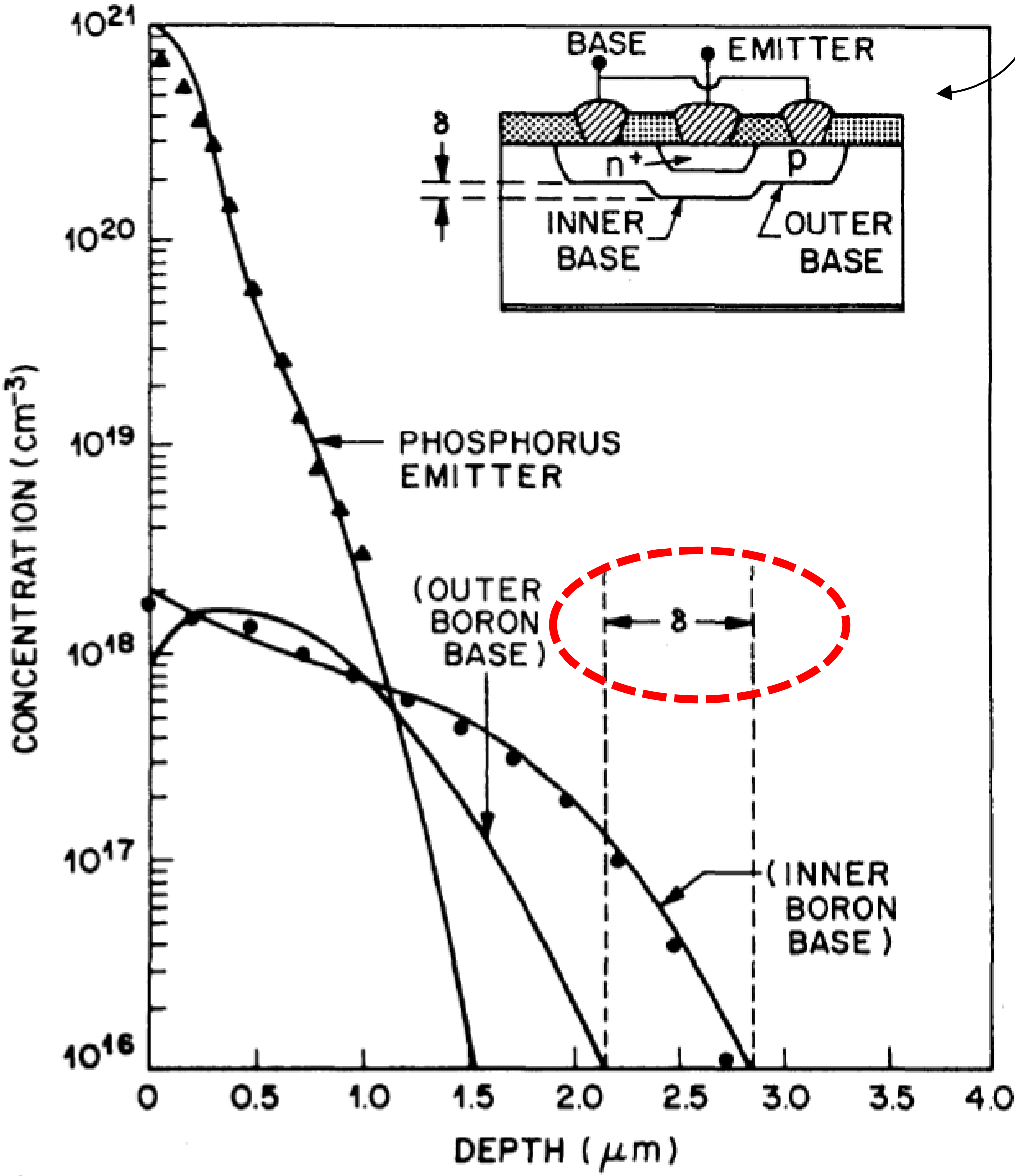
- Intrinsic diffusivity (low concentration)
    - $D_0 = 3.85\ \mathrm{cm^2/sec}$
    - $E_A = 3.66\ \ \mathrm{eV}$
- Extrinsic diffusivity (high concentration)
    - Region 1: high concentration region, P^+ and doubly charged vacancy pair (P^+V^{2-})
    - Region 2: kink region, the coupled P^+V^{2-} pairs dissociate to P^+, V^-, and electron.
    - Region 3: a larger amount singly charged V^- released enhance the P diffusion in the tail region.



Secondary Effects

- Emitter push effect
- Electric field enhanced diffusion
- Oxidation enhanced diffusion
- Lateral diffusion under oxide windows

Emitter push effect



Si n-p-n bipolar transistors using a phosphorus-diffused emitter and a boron-diffused base

- **Emitter push effect**: the base region under the emitter region (inner base) is deeper (by up to 0.6 μm) than that outside the emitter region (outer base).
- The dissociation of phosphorus vacancy (P^+V^{2-}) pairs at the kink region provides a mechanism for the enhanced diffusion of phosphorus in the tail region.
- The diffusivity of boron under the emitter region (inner base) is also enhanced by the dissociation of P^+V^{2-} pairs.

Electric field enhanced diffusion

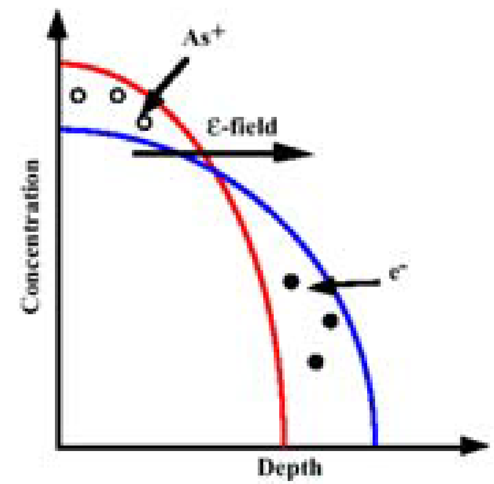
- E-filed is developed due to high mobility of electrons and holes in comparison with low mobility dopant ions.
- This internal electric field enhances the diffusivity of the ionized impurity atoms.
- The diffusion flux of dopants in an electric field is given by

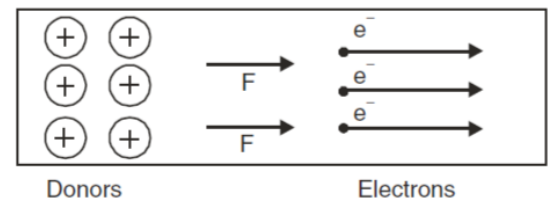
$$J = -qDh\left(\frac{\partial N_D}{\partial x}\right)$$

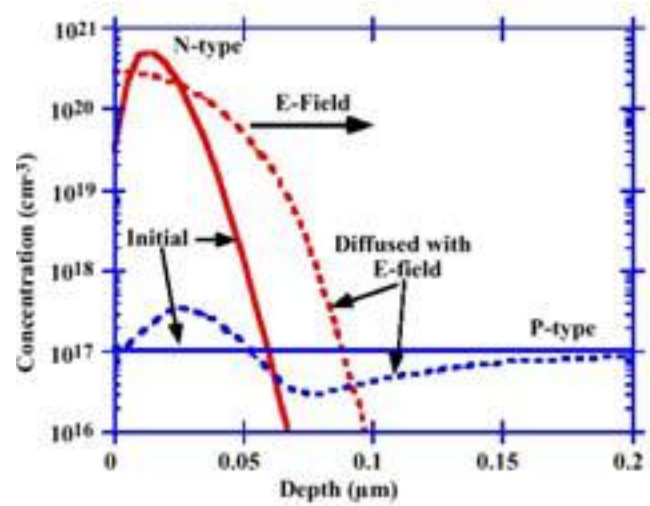
where h is the electric field enhancement factor given by

$$h = 1 + \frac{N_D / 2n_i}{\sqrt{(N_D / 2n_i)^2 + 1}}$$

- When $N_D >> n_i$, $h \simeq 2$. Thus, the maximum obtainable electric field enhancement of the diffusion constant is 2.

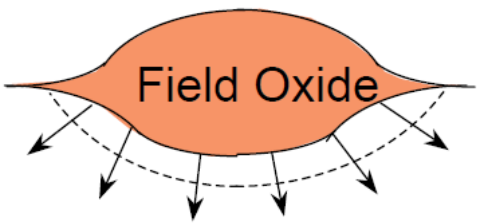


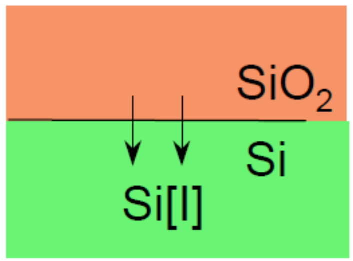


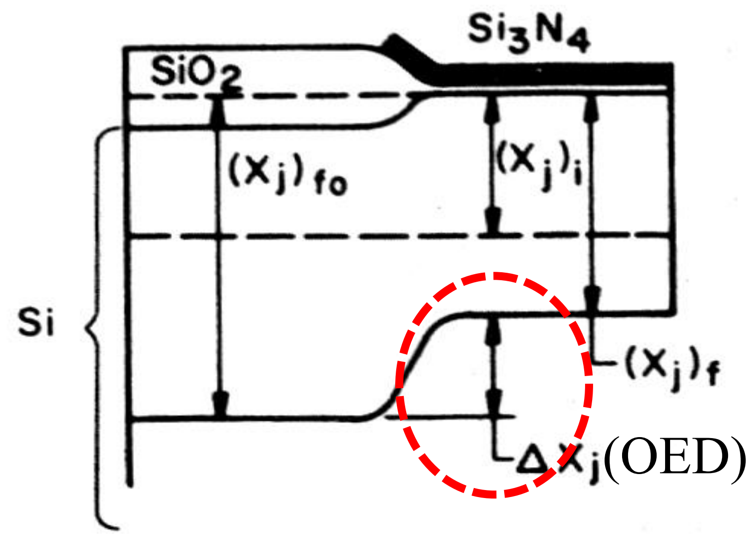


Oxidation Enhanced Diffusion (OED)

- During the thermal oxidation, **excess Si-interstitials** are generated at the oxidizing surface and diffuse into the bulk, resulting in **interstitial supersaturation** in the bulk.







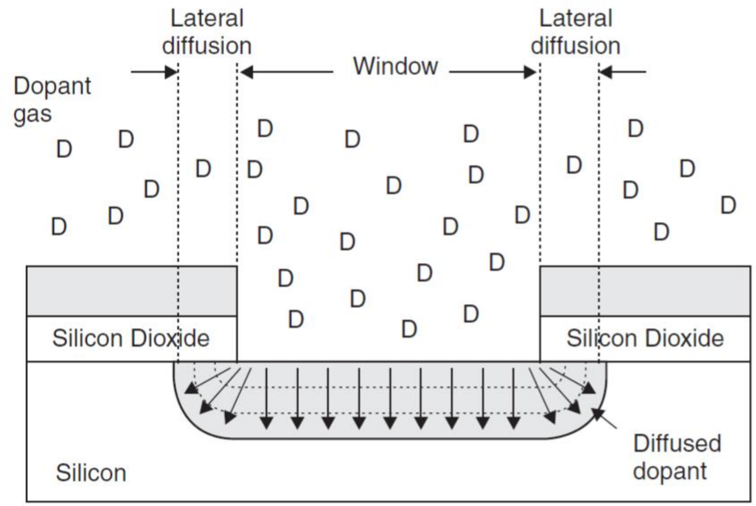
- This interstitial supersaturation causes an enhanced diffusion if the dopant diffuses via interstitial.

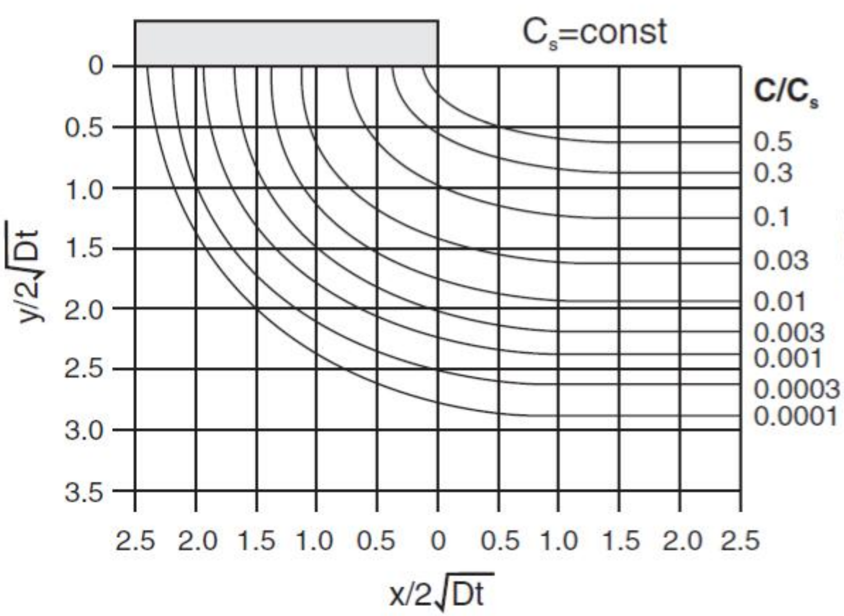
- Boron, Phosphorus : strong enhanced diffusion
  Arsenic : weak enhanced diffusion

- Boron & phosphorus have a large fraction (50%) of interstitialcy component
- Arsenic has a small fraction of interstitialcy component

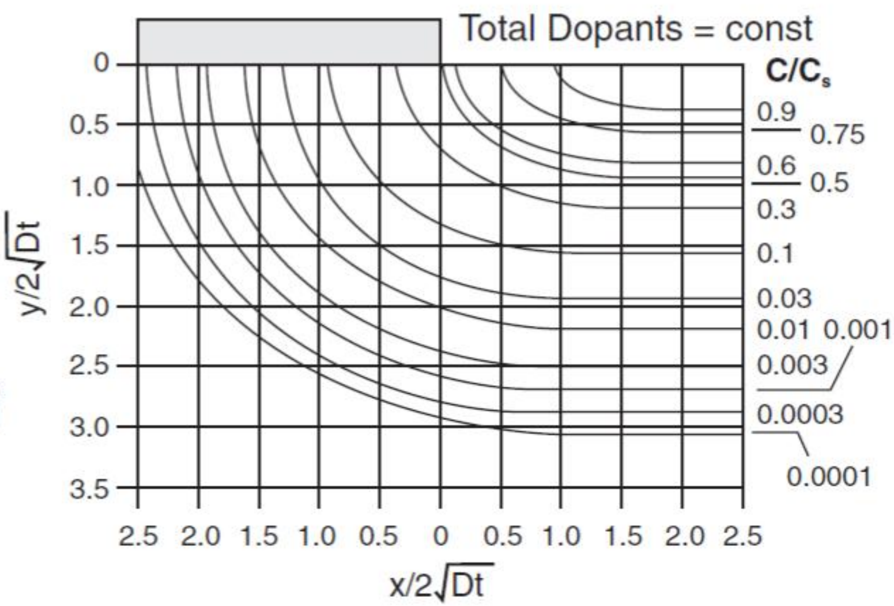
Lateral diffusion under oxide windows

- Lateral diffusion under the masking layer is an important parameter.
- In concentration-independent diffusion cases, the lateral penetration is 75 ~ 85 % of the vertical diffusion depth.
- In concentration-dependent diffusion cases, it is 65 ~ 70 % of the vertical diffusion depth.





a) Complementary error function

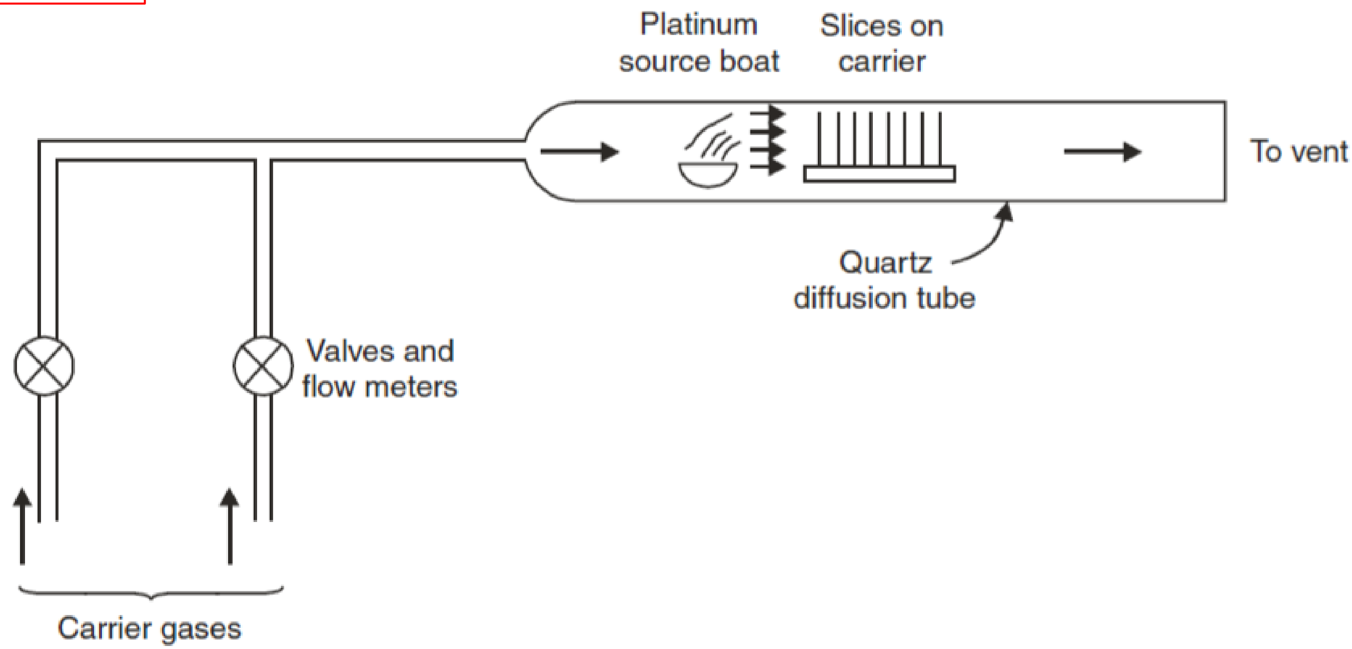


b) Gaussian

Diffusion System

- Diffusion systems typical use similar furnace or tube systems as oxidation process.
- Dopant delivery system
  - Solid source
  - Gas source
  - Liquid source

Solid source

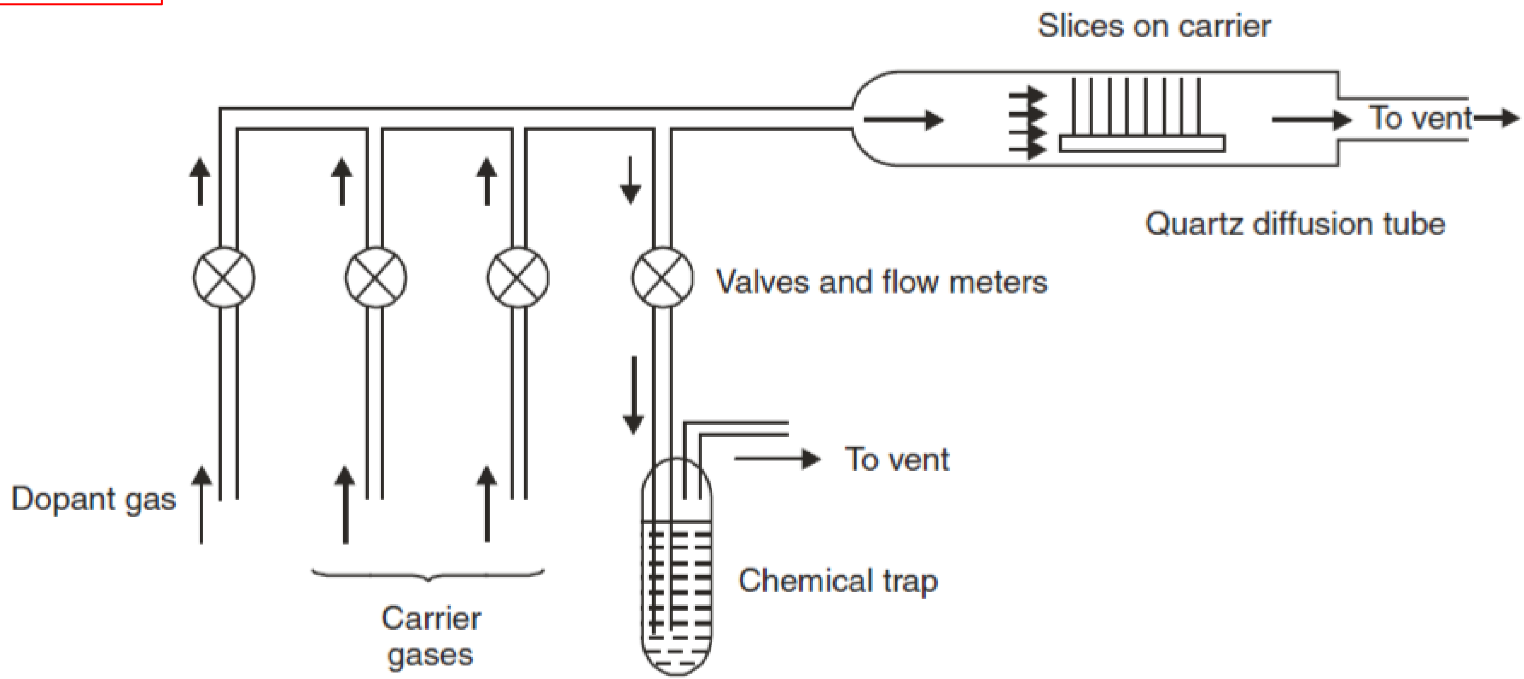


- Solid sources contain dopants in the form of B_2O_3 or P_2O_5.
- At high temperature, B_2O_3 diffusive into Si on carrier with the following chemical reaction:

$$2B_2O_3 + 3Si \rightarrow 4B + 3SiO_2$$

- B contained in SiO_2 (BSG) serves as B source during the drive-in annealing.

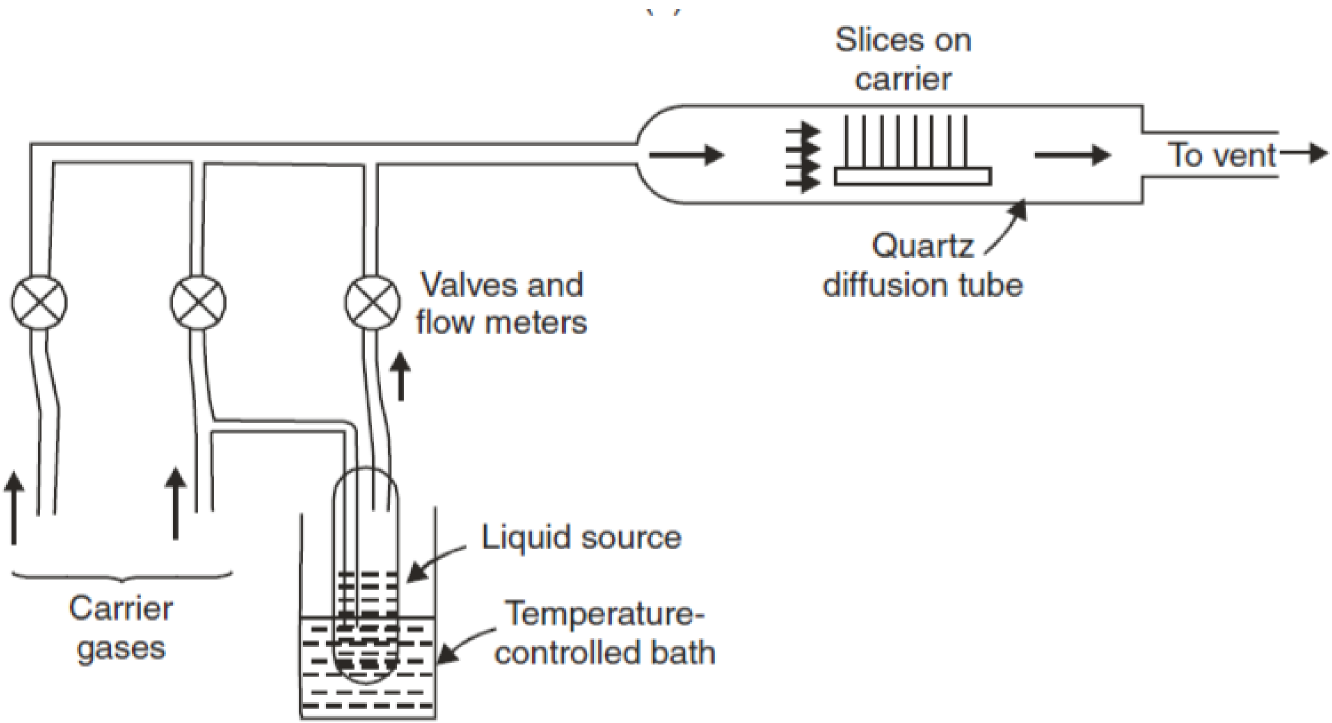
Gas source



- Common sources : PH_3, B_2H_6
- The source gas is mixed with a carrier or diluent gas (usually N_2) and a reactive gas such as O_2.
- The source gas reacts with oxygen at the wafer interface to form a dopant oxide (e.g. P_2O_5, B_2O_3).

$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$$

Liquid source



- Bubbler converts the liquid source to vapor by flowing a carrier gas.
- $POCl_3$ is the most widely used gas source for polysilicon doping.

$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$